CRIMINOLOGY

POLICING THROUGH SUBSIDIZED FIREPOWER: AN ASSESSMENT OF RATIONAL CHOICE AND MINORITY THREAT EXPLANATIONS OF POLICE PARTICIPATION IN THE 1033 PROGRAM*

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KEYWORDS: police, militarization, minority threat, economic threat, social control

There is a popular belief that the use of military equipment can improve police efforts at social control. Recent protests and riots across the country, however, have piqued public concern about racial disparities in law enforcement and the acquisition and use of military equipment by police in the United States. By using data from the Department of Defense's 1033 Program, we conduct an agency-level analysis to assess the validity of rational choice arguments and minority threat explanations of police participation in the 1033 Program. Our results reveal that higher violent crime rates and lower drug arrest rates increase law enforcement participation in the 1033 Program. Participation in the 1033 Program, however, is also a function of minority threat, with the functional form of minority threat varying across models predicting 1033 participation and the value of materiel acquired by successful departments. Specifically, a curvilinear relationship exists between the relative size of the Black population and involvement in the 1033 Program, and an exponential relationship between the relative size of the Black and Hispanic populations and the value of property departments receive annually from the 1033 Program.

Police use of force has been receiving increased public scrutiny in the United States recently. One example is the attention given law enforcement agencies (LEAs) acquiring military equipment, enabling police to deploy force with armored vehicles, aircraft, and automatic weapons (Balko, 2013). Criminologists and criminal justice scholars contend that the acquisition of such military gear; the growth of paramilitary police units (PPUs) or Special Weapons and Tactics (SWAT) teams; and the adoption of military terminology to describe domestic crime problems exemplify the militarization of law enforcement (Balko, 2013; Kraska, 2007). Indeed, domestic law enforcement has increasingly patterned its appearance and strategies after the military and military conflict (Kraska,

* Additional supporting information can be found in the listing for this article in the Wiley Online Library at http://onlinelibrary.wiley.com/doi/10.1111/crim.2018.56.issue-4/issuetoc.
Preliminary findings from this research were presented at the 2015 annual meeting of the American Society of Criminology in Washington, DC. We would like to thank Ryan King for helpful critiques on a previous draft.
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© 2018 American Society of Criminology doi: 10.1111/1745-9125.12193

2001). Although journalists, policy makers, and scholars are paying more attention to the potential consequences of police obtaining and using military gear, the reasons why departments acquire and use such equipment remain elusive.

To assess these patterns, we examine factors associated with participation in the Department of Defense's (DoD's) Defense Logistics Agency 1033 Program, one of the primary sources of military equipment for law enforcement (Dansky, 2014; Defense Logistics Agency, 2016a; Endebak, 2014; Grasso, 2014). In doing so, we draw from criminological and sociological theories of rational public choice and minority threat. Rational public choice theories posit that police acquire and use military equipment not only as a reaction to high crime rates but also in response to the wars on drugs and terrorism, which have changed the nature of policing (Bove and Gavrilova, 2017; Campbell and Campbell, 2009, 2016; Harris et al., 2017). A parallel argument contends that law enforcement priorities reflect the desires of politically and socially powerful majorities in society (Blalock, 1967), suggesting that the acquisition and use of military equipment is a response to growing minority populations viewed as threatening to the majority's social order (Ajilore, 2015; Balko, 2013; Meeks, 2006).

To date, research on the use of military equipment and tactics by police has been focused on a few national surveys, case studies, and ethnographic work; researchers have not explicitly addressed the role of racial and ethnic group relations in the acquisition of military equipment. Other research findings, however, have demonstrated a potential for social factors to influence police militarization by way of modern "tough-on-crime" initiatives, which disproportionally impact socially marginalized populations (Alexander, 2012; Balko, 2013; Beckett, 1999; Wacquant, 2009). Consequently, we draw from multiple criminological and sociological perspectives to conduct one of the first nationwide ecological studies on the acquisition of military equipment in the United States at the LEA level. To explore our research questions, we take advantage of yearly agency-level data on the value of military equipment obtained by police departments from the DoD's 1033 Program to assess varying explanations for the annual changes in the value of military equipment obtained by LEAs. Our results reveal that rational choice explanations, reflected in local violent crime rates and county drug arrests, partially account for law enforcement participation in the 1033 Program. Participation in the 1033 Program, however, is also a function of minority threat as the relative size of the Black and Hispanic populations predict involvement in the 1033 Program in a way that is consistent with minority threat expectations (Blalock, 1967). These findings for 1033 Program participation may indicate that police militarization is influenced by minority threat explanations independent of factors associated with rational public choice.

POLICE MILITARIZATION AND THE 1033 PROGRAM

Policing is often described as a quasi-military institution, relying on military ideals and trappings such as formal ranks, uniforms, discipline, structures, equipment, and culture (Bittner, 1970; Jermier and Berkes, 1979). Although criminologists and criminal justice scholars have debated the appropriateness or usefulness of using the military as a professional model for law enforcement (see Cowper, 2000; den Heyer, 2014; Waddington, 1987, 1993; cf. Buerger, 2000; Jefferson, 1993, 1987; Kappeler and Kraska, 2015), many American politicians, law enforcement professionals, and average citizens assume that the emulation of military traditions, tactics, chain of command, culture, and equipment

can supplement and improve police efficacy (Bove and Gavrilova, 2017; Campbell and Campbell, 2009, 2016; Harris et al., 2017; Phillips, 2016; Turner and Fox, 2017). To assess empirically how police may or may not be adopting or emulating such militaristic practices, however, as well as to assess the effect of such practices on police work, we must be clear about what is meant specifically by the term *police militarization.*

Peter Kraska defined militarization as a process of "arming, organizing, planning, training for, threatening, and sometimes implementing violent conflict" (2007: 503). Militarization itself is an extension of the underlying ideology of militarism, which is "a set of beliefs, values, and assumptions that stress the use of force and threat of violence as the most appropriate and efficacious means to solve problems" (Kraska, 2007: 503). Kraska (2007) also theorized four dimensions of militarization in police work: 1) the material use of military technology; 2) the cultural use of military language, values, and style; 3) the use of military organization principles to manage police work; and 4) the use of military tactics and principles to conduct police work. This four-dimensional framework is useful for scholars because "police militarization" has been written about in several synonymous terms (i.e., militarization or paramilitarization), yet applied across dissimilar concepts and activities within the law enforcement profession, such as organizational structures, culture, and management styles (Buerger, 2000; Cowper, 2000; den Heyer, 2014; Kappeler and Kraska, 2015; Kraska, 1996; Kunselman, Vito, and Walsh, 2013) and the acquisition and use of military equipment and tactics in police work (Bove and Gavrilova, 2017; Delehanty et al., 2017; Kraska and Kappeler, 1997; Harris et al., 2017; McQuoid and Vitt, 2017; Phillips, 2016; Phillips, Wheeler, and Kim, 2016; Radil, Dezzani, and McAden, 2017).

Even though Kraska's (2007) four-dimensional framework offers a comprehensive understanding of police militarization as a sociological phenomenon, we focus our study most heavily on the first dimension: the material use of military technology and equipment. Specifically, we explore the determinants of police acquisition of military-grade communications and transportation equipment, military fatigues, firearms, and ordinance preferred by modern militaries (i.e., materiel; Bove and Gavrilova, 2017; Bieler, 2016; Harris et al., 2017; Insler, McMurrey, and McQuoid, 2016; Kraska and Kappeler, 1997; Phillips, 2016). Our focus on this first dimension is motivated by the fact that such materiel makes LEAs more capable of participating (and dominating) in violent confrontations, which is a fundamental aspect of Kraska's (2007) conceptualization of militarization.

For LEAs in the United States, a popular source for acquiring such military equipment and technology is the DoD's 1033 Program (Dansky, 2014; Endebak, 2014; Grasso, 2014; Harris et al., 2017; Radil, Dezzani, and McAden, 2017). The 1033 Program originated with the National Defense Authorization Act of 1990 and, under several different titles, has provided LEAs military equipment warehoused to storage (for a maximum of 2 weeks) by the DoD. Although scholars and journalists have fixated on the combat equipment (including uniforms, body armor, firearms, radios, night-vision goggles, mine-resistant armored protected vehicles or MRAPs, and aircraft), the 1033 Program also provides materiel such as survival gear, medical utensils, and office furniture and supplies. Researchers have found that some LEAs may take advantage of office supplies and other noncombat materiel to offset policing costs (Bruce et al., 2017), whereas others may use combat materiel for law enforcement efforts (Bove and Gavrilova, 2017; Harris et al., 2017).

Even though there is a growing body of scholars who argue that participation in the 1033 Program makes an excellent test case for theories of police militarization (Ajilore, 2015; Bove and Gavrilova, 2017; Dansky, 2014; Delehanty et al., 2017; Harris et al., 2017; McQuoid and Vitt, 2017; Radil, Dezzani, and McAden, 2017), it is not an exhaustive indicator of police militarization.[1] We focus on the 1033 Program, however, because accessing materiel such as weaponry, vehicles, and uniforms is an early and often necessary step for many LEAs toward further militarization in the dimensions of language and terminology, organizing using military principles, and practicing military-style tactics (Ajilore, 2015; Insler, McMurray, and McQuoid, 2016; Kraska, 1996; Phillips, 2016). Moreover, the fact that the 1033 Program provides free materiel allows resource-poor LEAs the capacity to create PPUs and emulate policing efforts that may otherwise only occur in larger, better resourced departments (Dansky, 2014; Radil, Dezzani, and McAden, 2017).

The process for obtaining 1033 Program materiel is not difficult for LEAs and does not distinguish between combat and noncombat gear (Bruce et al., 2017; Harris et al., 2017). The first step is for the LEA to register with the 1033 Program by submitting a two-page form. Once an LEA is registered and decides to request materiel, the head administrator requests, via another form, specific materiel (e.g., firearms, armored vehicles, aircraft, or any other gear; Bruce et al., 2017; Defense Logistics Agency, 2016a; Harris et al., 2017). Other than the subordinate relationship that leaders of law enforcement have with local political leaders (e.g., mayors and local councils; Jacobs and O'Brien, 1998), there is little community oversight into the types of property LEAs acquire each year (Bruce et al., 2017).

Once a request is approved, the 1033 Program provides the requested equipment to LEAs (Defense Logistics Agency, 2016a; Grasso, 2014). The only financial cost to LEAs in this process is shipping charges. The availability of materiel, and its disposition to LEAs nationwide, is mainly determined by U.S. military engagements abroad and DoD decisions regarding the equipment that active military use when they are deployed (e.g., military drawdowns in Iraq and Afghanistan increased the availability of MRAPs for LEAs). Currently, the DoD does not evaluate the social control needs of an LEA's request for 1033 Program materiel, although the 1033 Program does prioritize requests that cite drug law enforcement or counterterrorism efforts and requires that departments plan to use the materiel for valid law enforcement efforts within the year.

THEORIES OF SOCIAL CONTROL AND THE POLICE

Although few dispute the fact that the police are increasingly adopting military practices and equipment, and that the 1033 Program may be influencing that trend, we know little about the determinants of 1033 Program acquisitions by LEAs. As such, to inform our assessment of militarization in the material dimension in the 1033 Program, we turn to

1. LEAs may use federal grant monies to purchase materiel from private vendors (Ackerman, 2014; Parlapiano, 2014). Such approaches, however, are subject to higher levels of oversight and are more expensive than the 1033 Program (Bove and Gavrilova, 2017; Department of Homeland Security, 2018). The 1033 Program is the primary source of certain materiel (e.g., MRAPs), and it provides fully automatic firearms, body armor, armored vehicles, and aircraft at significant value (Bove and Gavrilova, 2017; Feldman, 2014; Fowler, 2017; McRobbie, 2014; Welna, 2014). Moreover, LEAs must deploy 1033 property within a year (DLA, 2016b), indicating active militarized crime control.

theories of social control based on rational public choice and minority threat perspectives, which have informed prior work on police expenditures.

RATIONAL PUBLIC CHOICE

By drawing on econometric arguments for crime and crime control (Becker, 1968), rational public choice theories of social control presume that police departments will intensify social control efforts when facing greater demands to stop crime (Nagin, 1998; Stucky, 2005). As such, rational public choice theories posit that increases in policing efficacy (be it by expanding police forces, increasing funding, or improving overall competence) may increase the police's deterrent effect against crime and disorder (Cohen and Felson, 1979; Nagin, 1998).[2]

In this sense, police militarization may be seen as a rational public choice police response to social control demands that stem from high crime rates, participation in drug interdictions, threats to public safety, or even threats to the police themselves (Hall and Coyne, 2013). Indeed, Phillips (2016) argued that perceptions of lethal danger to police officers, and an institutional ethos of police as *crime fighters,* have become justifications for police to seek and carry military-style "patrol" rifles as a reasonable (yet unproven) response to criminal, terrorist, and mass shooter threats that have recently increased demands on LEAs for public security. Similarly, Kraska and Kappeler (1997) found that a primary justification for the existence of many paramilitary police units (PPUs, or SWAT units) is a belief that such gear and tactics improved LEA efforts at hot-spots policing, a policing strategy itself predicated on rational public choice theories of deterrence (Braga, Papachristos, and Hureau, 2014; Cohen and Felson, 1979; Roncek and Maier, 1991). Furthermore, Bell (1982: 47) argued that many police uniforms use military stylings to convey authority and power as general criminal deterrents, going so far as to say that police have "confused the desire to be military with the need to appear uniform and identifiable." Bell's (1982) insights are still relevant today as more and more LEAs may not simply add militaristic flair to their uniforms but fully adopt military fatigues and tactical vests used by the military to deter criminals (DeMichele and Kraska, 2001).

Following the rational public choice argument of policing, the acquisition of military equipment may serve as a means to improve LEAs' capacity to respond to crime or public safety crises. Indeed, recent research findings reveal that 1033 Program materiel acquisitions are associated with reductions in the rates of violent crime and property crime (Bove and Gavrilova, 2017) and with increases in drug arrest rates (Harris et al., 2017), although these findings are unclear on whether crime or arrest rates influence demand for 1033 Program materiel. If police militarization is indeed a response to increased demands on LEAs to produce social control, we anticipate that 1033 Program participation will be greater in places where police departments are called to respond to high violent crime rates, improve the arrest capabilities in of drug enforcement efforts, and effectively prepare for potential threats to public safety.

2. A complete review of this research literature is beyond the scope of this article; suffice it to say that we think it is important to account for a rational public choice dynamic as a determinant of police acquisition of military equipment. But interested readers should see Marvell and Moody, 1996; Stucky, 2005, 2013; and Stucky, Heimer, and Lang, 2007, for a review of rational public choice arguments in police expeditures and resource allocation.

MINORITY THREAT

Minority threat perspectives of social control are based on conflict theories predicated on racial/ethnic cleavages. Blalock (1967) theorized that increases in minority group presence create perceptions of racial competition with dominant majority groups. As a reaction to perceptions of competition for political and economic resources, the majority group may respond to minority group threats with harsher law enforcement practices (Liska, 1992). Later research findings on minority threat theories revealed that prejudiced attitudes in majority groups can indeed be associated with the presence of racial minorities (Giles and Buckner, 1993; Quillian, 1995, 1996; Taylor, 1998). In this way, crime control becomes a means for racial majorities to protect their status by pacifying or eliminating threatening minority groups (Behrens, Uggen, and Manza, 2003; Olzak, 1992; Spitzer, 1975).

When the percentage of Blacks, Hispanics, and non-White minorities in cities as an indicator of minority threat have been used, scholars in multiple studies have found a link between increased minority presence and larger police forces and police spending (Jackson, 1989; Jackson and Carroll, 1981; Kent and Jacobs, 2005; Liska, Lawrence, and Benson, 1981; Sharp, 2006; Stucky, 2005). In other studies, scholars have found that such minority threat explanations can also account for disparate minority incarceration rates (Bridges and Crutchfield, 1988; Garofalo, 1980), sentencing outcomes (Crawford, Chiricos, and Kleck, 1998; Johnson, 2005; Sampson and Laub, 1993; Ulmer and Johnson, 2004), and the use of extralegal violence against minorities (Chamlin, 1989; Holmes, 2000; Jacobs, Carmichael, and Kent, 2005; Jacobs and Wood, 1999; Sorensen, Marquart, and Brock, 1993). In short, the link between minority threat, the allocation of criminal justice resources, and the use of violent social control has emerged as a consistent finding in criminological research.

Although previous research findings have provided support to the notion that racial Whites are the dominant majority in the United States, there is some question about which racial minority group is the most threatening to this majority group and whether similar relationships between minority prevalence and social control practices exist across different minorities. Even though most studies of minority threat have been focused on Black populations (Jackson, 1989; Keen and Jacobs, 2009; King, 2008), others have produced results that reveal that Hispanic populations also produce threat responses from Whites (Kane, 2003; Kent and Jacobs, 2005; Stewart et al., 2015). It is theoretically plausible that any minority group can constitute a threat to a majority once it is large enough (Blalock, 1967); therefore, to assess minority threat explanations accurately, we must capture multiple measures of racial and ethnic minorities.

Scholars testing minority threat explanations often find the relationship between minority group threat and the allocation of criminal justice resources to be nonlinear. Blalock (1967) originally postulated that although perceptions of racial competition increase linearly with minority group presence, discriminatory responses against minority groups (such as disproportionate criminal justice resource allocation) would be nonlinear (Blalock, 1967: 140–7). According to Blalock (1967: 147–50), discriminatory actions cumulative in their effect to hinder minority competition would have a functional form that indicates a curvilinear relationship as the cumulative effect of discrimination to prevent threats reduced the need for harsher responses as minority presence grew. Discriminatory efforts, however, not cumulative in their effect to prevent minority competition would

motivate discriminatory responses that exponentially increased with minority group presence (Blalock, 1967: 1950–4).[3]

Although arguments for nonlinear responses have informed a broader conversation about the functional forms in minority threat and social control (Liska, 1992), researchers on minority threat and policing have regularly found nonlinear minority threat responses that vary in their functional form. Jacobs (1979) found that city-level police forces increase linearly with Black populations. Jackson and Carrol (1981), in their work on city-level police force spending, discovered that police spending increased until Black populations reached 50 percent and then declined for higher percentages. Kent and Jacobs (2005) and Kane (2003) likewise found a curvilinear relationship for police spending, increasing exponentially once Hispanic populations reached ~20 percent of the population in a locality. Furthermore, Stucky (2005) revealed a curvilinear association between police force size and both Black and Hispanic populations that was conditional upon the political context of city governments. Stults and Baumer (2007) similarly found county police force sizes increased at a decreasing rate until county populations reached 20 percent Black; then police force size began to decline. Given the wide range of findings between different minority groups and the variety of functional forms in criminal justice responses, in our assessment of 1033 Program participation, we consider functional forms of minority threat that might not be consistent across different racial and ethnic minority groups. It is also crucial, however, to understand how 1033 Program participation and the acquisition of militarized equipment may constitute a police response to minority threat, which leads us to explore the connection between race and police militarization.

1033 PROGRAM PARTICIPATION AS MINORITY THREAT RESPONSE

Although ecological associations have remained unexamined, previous research findings suggest that race is likely a factor in police militarization. In their seminal work on the expansion of police militarization, Kraska and Kappeler quoted an interview with a police commander who stated that the use of military equipment and tactics were meant to target "suspicious vehicles and people. We stop anything that moves. We'll sometimes even surround suspicious homes and bring out the MP5s. We usually don't have any problems with crack-heads" (1997: 10). The mention of using submachine guns (MP5s) to eliminate "problems" speaks to the intended symbolic threat of military gear in police work mentioned earlier, whereas the mention of suspicious vehicles can also have racial connotations (Kowalski and Lundman, 2007). At the same time, the mention of "crack-heads" belies the differential enforcement of drug laws against crack-cocaine users, who tend to be poor and Black, compared with powder-cocaine users, who tend to be wealthier and White (Alexander, 2012). Even more explicit in linking racial bias to police militarization, Kraska and Kappeler (1997: 10) later quoted a PPU commander, who stated that "when the soldiers ride in you should see those blacks scatter."

3. Blalock (1967: 147–54) argued that the former curvilinear form, increasing at a decreased rate, could be found in efforts to prevent minority economic competition even though the latter curvilinear form, increasing at an increased rate, could be found in efforts to ensure majority political dominance.

Although policy proponents contend that 1033 Program participation is a race-neutral response to violent crime and terrorism, pervasive stereotypes of minorities as criminogenic and threatening to Whites suggest to us that 1033 Program participation specifically, and police militarization generally, may be influenced by minority group presence (Balko, 2013; Bove and Gavrilova, 2017; Kraska, 2001). Meeks (2006) argued that police militarization constitutes an "obscure war against the urban underclass" motivated by shifting federal funds from social programs meant to aid minority communities into tough-on-crime programs, a trend occurring since the late 1960s (Hinton, 2016). These efforts to increase state social control in minority communities while diverting funds from social welfare programs to crime-control efforts are consistent with Wacquant's (2009) critiques. Thus, police have been able to justify the acquisition and use of military equipment and PPUs to conduct a variety of practices, from routine patrols to no-knock drug raids in minority communities defined as "problems" by police and politicians (Balko, 2013; Hinton, 2016; Kraska and Kappeler, 1997; Meeks, 2006; Phillips et al., 2016).

Perceptions of minorities as threatening to Whites, as well as law enforcement and policy maker responses to such threats, are well documented. Russell-Brown (1998) demonstrated that despite appeals of "color-blindness" from lawmakers and the police, White fears and stereotypes of criminal behavior remain dominated by racial biases that allow the proliferation of racial hoaxes like the "criminalblackman" stereotype. Indeed, popular perceptions of minority groups as criminogenic may influence policy leaders and law enforcement to view minority group presence as evidence of crime-control demand (Alexander, 2012; Du Bois, 1994 [1903]; Hinton, 2016; Weaver, 2007). Weaver (2007) also noted that the creation of the Law Enforcement Assistance Administration (LEAA) in 1968, the first federal initiative to provide federal funds and military equipment to local police, was in part a response to White perceptions of Black civil rights mobilization as criminal. Jackson and Carroll (1981), in their earlier work linking LEAA funding for police departments with a curvilinear minority threat response to Black population size and civil rights mobilization, further supported the idea that police militarization is minority threat response. Similarly, Hinton (2016) also noted that some of the earliest arguments for issuing military equipment to police—and some of the earliest efforts to issue the equipment—were motivated by White fears of urban Black rioters after the Watts and Detroit riots. In short, if we follow the argument that an increase in minority groups also increases the perception of threats to social order (Blalock, 1967), which in turn incites punitive responses from the majority (Jacobs and Helms, 1999; Quillian, 1995, 1996; Taylor, 1998), and that the use of military equipment allows police to respond with greater force (Kraska, 2007), then the link between police militarization and minority presence is plausible.

It is unknown whether minority presence may account for recent 1033 Program participation independent of the previously argued rational public choice explanations. Although a viable minority threat explanation of police militarization can be made based on previous research aimed at examining threats to the political status quo from the civil rights movement (Hinton, 2016; Jackson and Carroll, 1981; Weaver, 2007), Blalock (1967), in his original argument for nonlinear minority threat responses, suggests that perceived threats to political power should elicit discriminatory responses that increase at an increasing rate until a minority group's size reaches a point at which its political and economic power becomes sufficient to resist such discrimination efforts (Stucky, 2005; Stults and Baumer, 2007). The minority threat response to Black populations, however, may not

have the same functional form as for Hispanic populations, which White majorities may perceive as more of an economic than a political threat (Kane, 2003; Kent and Jacobs, 2005; Wacquant, 2009). As our study is the first to be aimed at examining minority threat explanations of police 1033 Program participation, it is prudent to assess the validity of nonlinear trends in 1033 Program participation and whether they vary across minority groups (i.e., increasing at an increasing rate for one minority group while perhaps increasing at a decreasing rate for another). Thus, we draw on the approach of previous research in which minority threat perspectives were used and for which a range of functional forms in minority threat responses in police militarization was tested (Jackson and Carroll, 1981; King and Wheelock, 2007; Stults and Baumer, 2007; Wang and Mears, 2010).

RESEARCH QUESTIONS

The earlier discussion of rational choice and minority threat explanations for police militarization motivates our research questions regarding local law enforcement's participation in the 1033 Program. First, do rational public choice theories of police strength explain participation in the 1033 Program? If so, we should anticipate that high crime rates, violent threats to police, and terrorist threats create demand for increased police security, predicating 1033 Program participation. Second, is minority threat a determinant of participation in the 1033 Program? Third, and relatedly, given that minority threats produce nonlinear responses from majority groups, if minority threat is a determinant of police militarization, what is the functional form of this relationship? In the following analyses, we assess these three research questions by examining both rational public choice and minority threat explanations that account for the likelihood of an LEA to participate in the 1033 Program and the value of materiel obtained by the participant LEAs. Finally, given that the 1033 Program can supplement police work with noncombat gear (Bruce et al., 2017), as well as combat materiel (Bove and Gavrilova, 2017; Harris et al., 2017), it is crucial for an assessment of 1033 Program participation as a test of police militarization to account for the qualitative nature of equipment obtained. Thus, we pose a fourth question: Do rational public choice and/or minority threat accounts of police miltiarization explain the aquistion of military equipment as opposed to nonmilitary equipment?

DATA AND METHOD

1033 PROGRAM DATA

Our primary source of data is the DoD's Defense Logistics Agency Law Enforcement Support Office (LESO) 1033 Program. The LESO was created in 1995 to work exclusively with local law enforcement and to improve the effectiveness of the program, which had been rarely used until then. The LESO records the value of the property supplied to LEAs as the value it had when obtained by the military. It also provides users with federal supply category and class names used by the military, itemized details using NATO Stock Numbers (NSN), and the date, location, and name of LEAs receiving military equipment. Thanks to the Freedom of Information Act, these 1033 Program data are publicly available (Defense Logistics Agency, 2016b). As discussed earlier, the 1033 Program provides a wide variety of materiel, combat and noncombat, with LEAs only having to cover the costs of shipping. We used LESO descriptions and federal supply categories and classes

in these data to code materiel received by LEAs as either militarized or nonmilitarized. We present a list of militarized and nonmilitarized materiel in appendices A and B.

Our conceptualization of militarized equipment is informed by Kraska's (2007: 503) definition of militarism, such that we identified militarized equipment sent to LEAs as equipment that stresses "the use of force and threat of violence" and may help LEAs to prepare for and implement violence as a solution to problems, as well as to dominate in any situation in which violence might arise. Examples of militarized property include munitions (e.g., firearms, explosives, and missile launchers); militarized transportation (e.g., armored trucks, MRAPs, airplanes, and helicopters); military communication or surveillance equipment (e.g., night-vision goggles and wide-range radio antennae); uniforms (e.g., body armor and specialized combat clothing); and miscellaneous items, including weapons-training guides, crowd-control equipment, and aftermarket armor for vehicles (a.k.a. up-armor kits).

We conceptualize nonmilitarized equipment as property received by agencies that does not facilitate the use or threat of violence or provide police officers with a significant tactical advantage in wielding violence. Examples of nonmilitarized property include office supplies and computers (e.g., hardware, software, furniture, and supplies); standard telecommunications equipment (e.g., fiber optics, radar, and video recording); clothing (e.g., socks and underwear); agricultural and industrial tools (e.g., pest control, hand tools, laboratory supplies, and generators); safety and rescue equipment (e.g., fire-fighting and hazardous spill and containment); and health and medical supplies (e.g., field medical kits, portable reagents, and IV fluids). Additionally, we consider items that agencies may use in conjunction with militarized materiel but that have universal purposes (e.g., aircraft maintenance equipment and engine accessories) to be nonmilitarized. Finally, we consider items that are primarily used in emergency or disaster response (e.g., water purification systems and ordinance disposal kits) to be nonmilitarized.

Although our conceptualization of militarized and nonmilitarized property is rooted in prior theory and based on the intended or original use of the materiel, we recognize that tactical and combat equipment can be implemented for nonmilitarized purposes (e.g., search and rescue). For example, police departments in Harris County, Texas, credit the MRAPs they received through the 1033 Program for assisting in search and rescue operations after Hurricane Harvey (Blakinger, 2017), and Miami Beach police argue they need MRAPs to drive through floods and help with evacuation and protection during potential mass shooting situations (Iannelli, 2014). Although these uses reflect larger trends in the changing responsibilities of police in general (Beck and Goldstein, 2017; Simon, 2007), the war on drugs and the ability to maintain a superior force in the threat of potential violence remain the primary initial reasons for acquiring property from the 1033 Program (Apuzzo, 2014; DLA, 2016b; Grasso, 2014; Iannelli, 2017). Finally, regardless of how departments use MRAPs and other militarized property, the acquisition of such materiel is associated with harsh and overbearing policing in both police practice (Delehanty et al., 2017) and public opinion (Carriere and Encinosa, 2017).

Although participation in the 1033 Program provides limited evidence of police militarization, our approach and data offer an in-depth assessment of the materiel LEAs are receiving via the 1033 Program, which may facilitate further militarization of LEAs, and the ecological determinants of such acquisitions. Because the 1033 Program provides materiel at low cost and with few restrictions, an examination of 1033 Program participation provides insight into police decision-making that is less influenced by

political and institutional factors than others forms of police militarization (Bove and Gavrilova, 2017; Harris et al., 2017). As such, our primary motivation in this study is to offer an unprecedented assessment of the ecological factors explaining the transfer of $5 billion in lethal and nonlethal materiel to local law enforcement at little cost, with little oversight, and amidst a growing amount of public criticism and concern.

After identifying the 1033 Program property as militarized or nonmilitarized and aggregating annual values for number and acquisition value for each agency, we merge the 1033 Program data to crime and arrest data from the Uniform Crime Reports (UCR) and LEOKA data using the state and agency name. We combine our agency-level data with data from the U.S. Census, American Community Survey, Congressional Quarterly Voting and Elections Collection, and the National Consortium for the Study of Terrorism and Responses to Terrorism (START) *Global Terrorism Database*, using the UCR Crosswalk and addresses provided to the UCR.

SAMPLE

Because the law enforcement responsibilities of county and state agencies differ greatly from state to state and serve overlapping geographies and populations as cities and towns, we exclude county and state agencies, as well as specialized forces (e.g., university campuses and school districts), and focus on local (cities and towns) law enforcement agencies, as defined in LEOKA. We then link each local law enforcement agency to the Census place that best reflects its jurisdiction. This includes all incorporated places (e.g., cities, towns outside of the Northeast and Midwest, villages, and boroughs); Census-designated places (e.g., towns/townships in the Northeast and Midwest, unincorporated places in New England City and Town Areas, Consolidated Cities, and unincorporated statistical places with a police force).[4] In most places, only one law enforcement agency operates and police chiefs and administrators are often required to work with, and answer to, local elected officials (e.g., mayors and city councilmembers) on law enforcement matters. In places where more than one local agency operates, typically one local department and multiple task forces or special units focus on different aspects of law enforcement (e.g., local liquor control).

Although data on military transfers are available dating back to the inception of the program in 1989, we reduce our analytic sample to the years between 1995 and 2016.[5] We wish to focus on the period since the creation of the LESO, which is responsible for managing the program and which works exclusively with law enforcement (DLA, 2016b), to separate the limited scope of the program during its earliest years from more recent trends in police militarization as a general approach to policing (Balko, 2013). Although the amount of property increased substantially during the wars in Afghanistan and Iraq, we include prewar years (1995–2002) to capture the period in which Kraska and colleagues first identified trends in police militarization in the criminological literature (Balko, 2013; Kraska, 1996; Kraska and Cubellis, 1997; Kraska and Kappeler, 1997) and to account for early adopters of police militarization through the 1033 Program.

4. To save space, we use the term "place" to refer to all cities and towns.
5. Because we use 1-year lags, all values for our independent and control variables are measured between 1994 and 2015.

We exclude several LEAs for varying reasons. Because of their unique geographic and racial/ethnic contexts, we exclude Alaskan and Hawaiian agencies from all analyses. Because of the unique relationship between the federal government and D.C. law enforcement, we exclude the District of Columbia PD and DC Metro Police. After excluding agencies serving extremely small places (maximum yearly population between 1994 and 2015 of less than 500) and those agencies that never reported to the UCR between 1994 and 2015, our final sample size consists of 238,013 department-years' worth of data from 11,764 local police departments.

DEPENDENT VARIABLES

Our dependent variables capture the annual value (as recorded by the LESO) of militarized and nonmilitarized equipment per 10,000 persons (place population) obtained by individual local law enforcement agencies from the 1033 Program between 1995 and 2016. This measure allows us to account for the decision to participate in the 1033 Program, the degree to which LEAs are adding to their inventories, and the variation in types of military equipment that are available from year to year.

INDEPENDENT VARIABLES

Rational Public Choice Explanations

The predominate conventional explanations for police militarization specifically, and for aggressive policing generally, have been high rates of crime, risks to officer safety, and the changing role of domestic law enforcement in addressing both domestic and foreign terrorist threats. In particular, drug enforcement efforts, and violent crime associated with such efforts, remain the most frequently cited reasons given by law enforcement to explain aggressive policies (Bove and Gavrilova, 2017; den Heyer, 2014; Harris et al., 2017; Kappeler and Kraska, 2015). For example, police departments that report high rates of violent crime may need superior weapons, armored vehicles, and better surveillance and communications equipment to protect officers and detain criminals (Bove and Gavrilova, 2017; Harris et al., 2017). We measure *violent crime rates* using the number of homicides, aggravated assaults, robberies, and forcible rapes per 10,000 place population.

Although the 1033 Program made materiel available for efforts outside of drug interdiction in 1997, more than half of the requests from departments are for drug enforcement (Bove and Gavrilova, 2017; DLA, 2016b). Departments in counties with lower levels of drug arrests may view their enforcement efforts as ineffective, motivating them to participate in the 1033 Program for materiel that might improve their ability to detect and arrest criminals in the war on drugs. We include a measure of *drug arrest rates* using the county-level number of drug arrests (possession and sales) per 10,000 county population.[6] We choose a county-level measure of drug arrests task forces, as opposed to local arrests, for several reasons. First, buoyed by the recognition that drug dealers do not recognize

6. We note that arrest rates reflect police behavior (arrest), whereas crime rates include those that ended in arrest and those that were reported to the police but did not end in arrest. We chose arrest rates for two important reasons. First, there is limited data on drug usage and trafficking below the state level. Second, even if we had access to longitudinal local data on drug use, we argue that arrest rates still provide a better picture of local law enforcement's perceptions of drug crime, which then motivates rational choice responses such as acquiring or not acquiring military gear.

jurisdictional boundaries and that agencies needed to pool collective resources during the height of the drug war, many agencies now dedicate significant resources to multijurisdictional drug task forces, typically at the county or regional level (Jefferis et al., 1998; Smith et al., 2000). Second, although we are able to collect data from several state and local official sources to supplement missing data in the UCR, multiple geographies do not report drug arrest data at the agency level (e.g., Florida and Illinois) but do provide county-level estimates.

In addition to fighting violent crime and the war on drugs, police departments are now also expected to engage in counter-terror efforts. Departments in places that experience terrorism may be more likely to participate in the 1033 Program than departments in other places. We measure *recent terror attack* with a dummy variable equal to 1 for departments in places that have experienced at least one terror attack in the past 3 years using data from the START Global Terrorism Database (National Consortium for the Study of Terrorism and Responses to Terrorism, 2015).[7] Finally, the LESO provides access to vehicles and equipment that offer police more protection than common police cruisers and weapons. As a result, departments that have experienced police fatalities may be more likely to participate in the 1033 Program. We use LEOKA data to measure *recent officer fatalities* using the total number of officers killed on duty (felony and nonfelony) over the past 3 years per 100 officers.[8]

Minority Threat Explanations

We measure minority threat using data from the 1990 and 2000 U.S. Census, and because the American Community Survey replaced the long-form version of the decennial census, we use the 2005–2009, 2006–2010, 2007–2011, 2008–2012, 2009–2013, 2010–2014, 2011–2015, and 2012–2016 five-year ACS samples to construct our minority threat measures.

We test for different nonlinear relationships between racial and ethnic composition and the likelihood of participation in the 1033 Program as well as the value of materiel acquired by participating departments (Blalock, 1967; King and Wheelock, 2007; Stults and Baumer, 2007; Wang and Mears, 2010). We anticipate that racial and ethnic threat has a curvilinear relationship with participating in the 1033 Program and acquiring militarized property, but this relationship may be quadratic or logarithmic. As a result, we assess linear, quadratic, and natural logarithmic forms for percent Black (*percent Black)* and percent Hispanic (*percent Hispanic*) in our probit models predicting the receipt of any militarized property from the 1033 Program and our ordinary least-squares (OLS)

7. Terror attacks are rare occurrences that have long-lasting effects on local communities. Additionally, equipment may not be readily available or LEAs may take time to develop plans to prepare or respond to future attacks. As a result, we use a dummy variable capturing events within the past 3 years. Results do not vary if we measure terror attacks within the past year, 5 years, or a cumulative measure of all terror attacks including in the START data.

8. We use rolling 3-year averages for officer fatalities instead of annual averages because, similar to terror attacks, officer fatalities are rare occurrences that leave long-lasting effects on local communities, and we want to account for any time it might take for LEAs to respond to officer deaths by acquiring property from the 1033 Program. Results do not vary if we use rolling yearly or 5-year averages.

models predicting the value of militarized and nonmilitarized property departments acquired from the 1033 Program in a given year.[9]

Control Variables

We include several measures that could explain participation in the 1033 Program and the acquisition of high-value property at a discounted cost. Several scholars have contrasted racialized crime theories such as racial threat with conflict theories rooted in neo-Marxian perspectives of power and crime control (Rusche and Kircshheimer, 1939; Spitzer, 1975; Turk, 1969). Generally speaking, these arguments hold that large populations of the poor and unemployed present significant threats to economic and political elites, who use their influence to address these threats with criminal justice practices like policing (Carmichael and Kent, 2014; Kent and Jacobs, 2005; Stucky, 2005; Stucky, Heimer, and Lang, 2007). We use two variables that capture different aspects of economic conflict that may affect the ability of local law enforcement to access military gear through the 1033 Program. We measure the *poverty rate* as the percentage of the relevant jurisdiction living under the poverty line. We measure the *unemployment rate* as the percentage of the relevant jurisdiction that is out of work.

Because larger police departments may need more materiel for their officers, we include a measure of *police officers per 10,000 population*. Research findings show that politicians and localities supportive of the Republican Party tend to support increases in police expenditures, corrections expenditures, and punitive criminal justice polices (Beckett, 1999; Jacobs and Carmichael, 2002; Jacobs and Helms, 1999). Since police militarization operates to make law enforcement more punitive and coercive, we anticipate that localities supportive of the Republican Party may be inclined to pursue police militarization. Using data from the Congressional Quarterly Voting and Elections Collection, we measure political attitudes using the percentage of the county voting for the most recent Republican presidential candidate and interpolated for nonelection years (*percent Republican*). We control for regional differences in policing and access to military equipment using dummy variables for Midwest (reference), Northeast, South, and West Census regions, and for *population size* using the natural log to account for skew and nonlinearity.

We include several controls that capture the practical restraints that LEAs face when participating in the 1033 Program. First, we include a measure of *average distance to DLA disposition centers*. Because departments are required to cover the costs to inspect and transport property from DLA disposition centers to their agencies, we assume that departments closer to these centers will be more likely to acquire militarized property. We also recognize that if departments are seeking out a specific item, they may be willing to pay to ship it long distances. Relatedly, if departments are shipping from a long distance, they may request and acquire property of a greater value to maximize their shipping costs. We use the natural log of our distance variable to account for skew and nonlinearity. We control for *land area* (in logged square kilometers) because the 1033 Program gives priority to departments serving larger jurisdictions in terms of land, particularly for vehicles. We include a measure of *lagged total property value* to control for recent 1033 Program

9. Although our discussion revolves around the results from our best-fit models, we present results for all of our tests of nonlinearity in supplemental tables in appendices E through G in the online supporting information.

participation. Finally, we include a series of year dummy variables to control for changes in military engagements and annual fluctuations in the availability of materiel. Although some correlation among several of our predictors raises the possibility of problems related to multicollinearity (see appendix C and appendix D in the online supporting information[10]), the results of tests of variance inflation and sensitivity analyses revealed that multicollinearity is not an issue for any neighborhood type.[11]

MISSING DATA

There are two sources of missing data in our sample. First, because we use Census and ACS data for our demographic and social variables, we do not have measures for intercensal years (1997–1999, 2001–2008). We address this concern using linear interpolation to calculate year-by-year measures between the Census and ACS data points. This approach constructs new data points for observations between two or more known data points and is common strategy for handling the years between the Census and ACS in demographic (Crowder, Pais, and South, 2012; Ludwig et al., 2012) and criminological research (Eason, Zucker, and Wildeman, 2017; Parker, Mancik, and Stansfield, 2017; Xie and McDowall, 2014).[12] The second source of missing data is agency nonparticipation in UCR or LEOKA. After removing agencies that we could not locate any data on for the entire study period (1995–2015), slightly more than 6 percent of our agency-years contain missing data on either violent crime or drug arrests. We employ multiple imputation techniques to estimate these missing values using the "ICE" command in Stata 15.0 (Royston, 2005). ICE uses a chained equation approach, which produces a conditional distribution for missing data using the proper specification for each variable (e.g., OLS regression for continuous variables). ICE then creates multiple data sets using Gibbs sampling techniques (Royston, 2005; van Buuren, 2012). Prior research findings reveal that the number of data sets be at least equal to the percentage of cases with missing data (Carll, 2017; White, Royston, and Wood, 2009). We create ten unique data sets, which we then combine using the MI command in Stata 15.0 to complete descriptive analyses and multivariate analyses.

ANALYTIC STRATEGY

An overwhelming majority of LEAs do not participate in the 1033 Program in any given year and report receiving zero dollars' worth of property. Therefore, our dependent variable (value of property transferred) is clustered at zero and property values are only

10. Additional supporting information can be found in the listing for this article in the Wiley Online Library at http://onlinelibrary.wiley.com/doi/10.1111/crim.2018.56.issue-4/issuetoc.
11. Most notably, correlations between the linear and quadratic terms (e.g., percent Black and percent Black squared) were extremely high. Additionally, poverty and unemployment were highly correlated ($r = .58$). In each case, both predictors are conceptually important for this article, so we chose to keep them in our final models. Additionally, we tested variance inflation factors (VIFs) running separate regressions for our total sample and participating LEAs in Stata 15.0, clustering the standard errors at the county level, and found that the quadratic terms were the only variables to have variance inflation factors above 4.0.
12. To examine how the use of interpolated data might impact the study results, we ran supplemental results using 5-year intervals. Our results did not differ from those presented here. Because of fluctuations in the availability of 1033 property and the year-to-year differences in perceived social control needs of local law enforcement, we use interpolated data in all analyses.

observed when departments are acquiring materiel, violating multiple OLS assumptions. As a result, we employ Cragg double-hurdle models (with standard errors clustered at the county level to account for possible nonindependence among local LEAs) using Stata 15.0 to account for the clustering limit of the dependent variable at zero and the possibility that different factors explain acquiring property from the 1033 Program and the value of property acquired from the 1033 Program. The double-hurdle model fits a probit model predicting acquiring any equipment *for all departments* and an OLS model predicting the value of property acquired *for only participating departments*. We assume that zeros in the 1033 Program data are fundamentally different from nonzero values in that they represent a department not participating in the 1033 Program and, therefore, that these nonzero values constitute the first step in the process of participating in the 1033 Program in a given year; the second step in the process, decisions about the property the LEA acquires and will eventually use in a given year, is reflected in the per-capita value of that property. Please note that we chose hurdle models over other, similar models for limited or censored dependent variables (e.g., Tobit regression or Heckman selection models), for both empirical and conceptual reasons.[13] Finally, because we are interested in the degree to which *all* departments in the United States are participating (or not participating) in the 1033 Program, we use the *margins* command in Stata 15.0 to estimate unconditional marginal effects for all variables in the models and conditional marginal effects of percent Black, percent Hispanic, and our crime variables at their national medians, 75th, and 90th percentiles, respectively, while holding all other variables constant.

RESULTS

Table 1 presents descriptive statistics for total, militarized, and nonmilitarized property reported by total sample, for currently participating (e.g., values in the year departments participated) and for those police departments that participated in 1033 Program at least once during the study period. Between 1995 and 2016, the U.S. Department of Defense provided slightly more than 48,000 dollars of total supplies per agency nationwide. Although the total cumulative value of militarized property ($26,754) exceeded that of nonmilitarized property ($18,432), the per-capita value of nonmilitarized gear ($1,994 per 10,000 persons) was worth almost twice that of militarized gear ($1,002 per 10,000 persons). Among agencies that participated at least once in the 1033 Program between 1995 and 2016, the average cumulative value of total property transferred is slightly more than $128,000, including $75,791 of militarized property and $52,280 of nonmilitarized property. Although year-to-year participation is rare, the average annual amount of militarized property transferred is quite large. Each year, participating departments receive almost $80,000 per 10,000 persons in total property from the 1033 Program, including $26,513 of militarized property and $52,729 of nonmilitarized property.

Table 2 presents means and standard deviations for the independent and control variables in the total sample, as well as a comparison between those police departments that

13. Tobit regression assumes variables predicting censored and noncensored observations are the same (Bushway, Johnson, and Slocum, 2009; Smith and Brame, 2003), an assumption not reflective of our 1033 Program data. Likewise, Heckman models assume zero values are a latent, unobserved, or partially observed value of y (Bähr and Abraham, 2016; Bushway, Johnson, and Slocum, 2009), which is also not representative of zero values in our data.

Table 1. Descriptive Statistics for Dependent Variables Used in Analyses

	Total Agencies		Currently Participating Agencies		Ever Participating Agencies	
Variables	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**
Total Property						
Cumulative (dollars)	45,184.66	468.63			128,070.10	1,280.77
Annual (dollars)	1,933.16	68.44	17,041.30	1,219.64	5,478.97	193.52
Annual (dollars per 10,000 people)	2,997.10	154.16	79,243.48	3,993.07	8,501.06	436.63
Total Militarized Property						
Cumulative (dollars)	26,754.24	315.65			75,791.63	870.21
Annual (dollars)	1,234.94	59.43	10,536.11	1,016.21	3,498.52	168.29
Annual (dollars per 10,000 people)	1,002.80	99.63	26,513.58	2,620.16	2,844.35	282.50
Total Nonmilitarized Property						
Cumulative (dollars)	18,432.62	249.22	97,901.70	3,540.98	52,280.92	691.83
Annual (dollars)	698.23	28.37	6,505.19	544.90	1,980.47	80.28
Annual (dollars per 10,000 people)	1,994.32	105.09	52,729.90	2,724.80	5,656.75	297.67
N (Agency-Years)	238,013		83,913		9,002	
N (Agencies)	11,764		4,037		4,037	
N (Counties)	2,746		1,681		1,681	

Table 2. Descriptive Statistics for Independent and Control Variables Used in Analyses

	Total Agencies		Ever Participating Agencies		Never Participating Agencies	
Variables	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**
Minority Threat						
% Non-Hispanic Black	9.002	.034	**9.728**	**.057**	**8.606**	**.043**
% Hispanic	7.834	.030	**8.910**	**.053**	**7.247**	**.035**
Rational Choice and Public Safety						
Violent crimes per 10,000[a]	165.069	.505	**189.007**	**.809**	**150.090**	**241.120**
Drug arrests per 10,000[b]	47.562	.065	**49.096**	**.103**	**44.853**	**30.876**
Recent officer fatalities	.032		.032		.032	
Fatal terrorist attack	.004		.006		.002	
Control Variables						
% persons below poverty	14.283	.019	**14.606**	**.032**	**14.107**	**.024**
Unemployment rate	7.103	.010	**7.237**	**.016**	**7.030**	**.013**
Police officers per 10,000 population	2.110	19.227	2.040	18.292	20.150	19.738
Mean distance from DLAs	1,78.380	332.208	1,767.610	337.723	1,787.380	328.940
Land area (1,000 km^2)	14,200	52,300	21,500	69,700	**10,232**	**39,102**
Population	30,824	514	**52,994**	**1,381**	**18.752**	**.251**
% Republican	51.560	13.472	51.580	13.531	51.55	13.440
Region						
Midwest (reference)	.322		**.363**		**.300**	
Northeast	.236		**.160**		**.278**	
South	.327		**.352**		**.313**	
West	.114		**.125**		**.109**	
N (Agency-Years)	238,013		83,913		154,100	
N (Agencies)	11,764		4,037		7,727	
N (Counties)	2,746		1,681		1,065	

NOTE: Bold values indicate statistically significant differences between ever and never participating agencies ($p < .05$).
ABBREVIATION: DLA = Defense Logistics Age.
[a]City/town level.
[b]County level.

participated in the 1033 Program at least once during the study period and those that never participated in the 1033 Program between 1995 and 2016. Looking first at the minority threat variables, agencies that have participated in the 1033 Program represent places with significantly higher Black populations (9.7 percent versus 8.6 percent) and Hispanic populations (8.9 percent versus 7.2 percent) than agencies that have not participated. Although these numbers are slightly smaller than the national average for this period, the small mean values reflect the large number of small places in the data set and increasing residential "macro-segregation" in which Black–White and Hispanic–White segregation within places (e.g., big cities) has declined, but segregation between places has increased significantly over the past two decades as affluent Whites cluster in ex-urban communities, offsetting the increasing diversity of the central cities (Lichter, Parisi, and Taquino, 2015). Recent studies of metro and nonmetro Census places have yielded similar place-level percentages of Black and Hispanic populations (Lichter, Parisi, and Taquino, 2015; Parisi, Lichter, and Taquino, 2015).

We turn next to rational choice explanations of 1033 Program participation. On average, local police departments in the United States are cities and towns with an average of 165 violent crimes per 10,000 population and counties with 47.5 drug arrests (sales and possession) per 10,000 county population. These numbers are similar to prior research in which the UCR over the same time period was used (Beck and Goldstein, 2017; Feldmeyer, Harris, and Lai, 2016). When we compare the violent crime rates for participating and nonparticipating departments, violent crime rates are significantly higher in agencies that have participated in the 1033 Program (189 per 10,000 population) than for agencies that have not participated in the 1033 Program (150 per 10,000 population). Similarly, the county drug arrest rates for agencies that have participated in the 1033 Program (49 per 10,000 population) are slightly higher than the county drug arrest for agencies that have not participated in the 1033 Program (45 per 10,000 population). As police fatalities and terror attacks are extremely rare events, there are minimal differences in officer deaths or terror attacks across departments that participated in the 1033 Program and those that did not.

There are significant differences in our control variables as well. Compared with never participated departments, those that participated at least once have significantly higher poverty and unemployment rates, and have served larger populations and geographies. Notably, there is little difference between participating and nonparticipating agencies in terms of officers per capita, distance to DLA disposition centers, or proportion of Republican voters in participating places which is similar to the total sample.

The results from table 1 and table 2 provide early support for rational choice and racial threat arguments for 1033 Program participation. Specifically, agencies that militarize through the 1033 Program have higher rates of violent crime, as well as greater proportions of Black and Hispanic residents in their jurisdictions. To examine further these relationships over time in a multivariate model, table 3 presents results from double-hurdle models of 1033 Program participation.

Table 3 presents the results of probit models predicting *acquiring property* and OLS models predicting *property value*. We use the *margins* command in Stata 15.0 to estimate the unconditional average marginal effects (UAMEs) of our independent and control variables on property value for all departments annually. The results from table 3 indicate that both rational choice public safety factors and racial threat dynamics are associated with involvement in the 1033 Program. When we first turn to conventional rational

Table 3. Results From Double-Hurdle Model Estimating 1033 Participation via Acquiring Property and Value of Property Acquired for Militarized and Nonmilitarized Property, 1995–2016

Militarized Property	Acquiring Property	SE	Property Value	SE	UAME	SE
Minority Threat						
% Black	.003*	.002			.008*	.004
% Black-squared	.000**	.000			.000**	.000
% Hispanic	–.002	.002			–.005	.004
% Hispanic-squared	.000	.000			.000	.000
% Black (logged)			.070**	.023	.096**	.032
% Hispanic (logged)			.075*	.032	.111*	.048
Rational Choice and Public Safety						
Violent crime rates[a] (logged)	.030***	.005	.018	.012	.013**	.004
Drug arrest rates[b] (logged)	–.020*	.010	–.064*	.029	.000	.002
Recent officer fatality	–.025	.050	.145	.122	.001	.001
Fatal terrorist attack	.005	.081	–.071	.250	–.058	.323
Control Variables						
% persons below poverty	–.002	.001	–.005	.004	–.009	.005
Unemployment rate	.004*	.002	.008	.006	.018*	.007
Police officers per 10,000 (standardized)	.037***	.008	.103**	.033	.188***	.037
Mean distance from DLAs (logged)	–.074	.128	.324	.279	.000	.001
Land area (km^2) (logged)	–.001	.002	.009	.007	.000	.001
Population	.121***	.007	–.523***	.021	–.029***	.008
% Republican	.001	.001	.002	.002	.003	.003
Region						
Midwest (reference)						
Northeast	–.382***	.043	–.179	.115	–1.114***	.153
South	–.034	.028	.191*	.074	.118	.100
West	–.041	.058	.437**	.152	.346	.178
Lagged Total Property Value (logged)	.079***	.003	–.011	.010	.034***	.003
Intercept	–2.120***		6.731***			

Nonmilitarized Property	Acquiring Property	SE	Property Value	SE	UAME	SE
Minority Threat						
% Black	.001	.002			.003	.007
% Black-squared	.000	.000			.000	.000
% Hispanic	.002	.002			.007	.007
% Hispanic-squared	.000	.000			.000	.000

(*Continued*)

Table 3. Continued

Militarized Property	Acquiring Property	SE	Property Value	SE	UAME	SE
% Black (logged)			.024	.040	.033	.055
% Hispanic (logged)			–.032	.040	–.048	.059
Rational Choice and Public Safety						
Violent crime rates[a] (logged)	.058***	.008	.011	.024	.040**	.012
Drug arrest rates[b] (logged)	–.033	.019	.014	.044	.001	.002
Recent officer fatalities	–.033	.069	.025	.265	–.001	.002
Fatal terrorist attack	.254	.175	–.129	.273	.694	.561
Control Variables						
% persons below poverty	–.001	.002	.003	.005	.000	.008
Unemployment rate	.010***	.002	.018*	.008	.050***	.011
Police officers per 10,000 population (logged)	.046***	.010	.179***	.048	.328***	.059
Mean distance from DLAs (logged)	–.637***	.176	–1.260**	.411	.001	.019
Land area (sq. miles) (logged)	.008*	.003	.009	.013	–.002	.002
Population	.089***	.009	–.864***	.030	–.025	.019
% Republican	–.001	.001	.004	.003	.001	.006
Region						
Midwest (reference)						
Northeast	.170*	.073	.787***	.217	1.345***	.366
South	.196***	.041	.288*	.116	.930***	.175
West	.285**	.084	.473*	.211	1.395***	.339
Lagged Total Property Value (logged)	.088***	.003	–.010	.009	.049***	.004
Intercept	–4.055***		11.525***			
N (Agency-Years)	238,013		7,252		3,184	
N (Agencies)	11,764		4,037		1,869	
N (Counties)	2,746		1,681		932	

ABBREVIATIONS: SE = standard error; UAME = unconditional average marginal effect; DLA = Defense Logistics Agency.
NOTE: All models include year fixed-effects.
[a]Rates per 10,000 population (city or town).
[b]Rates per 10,000 population (county).

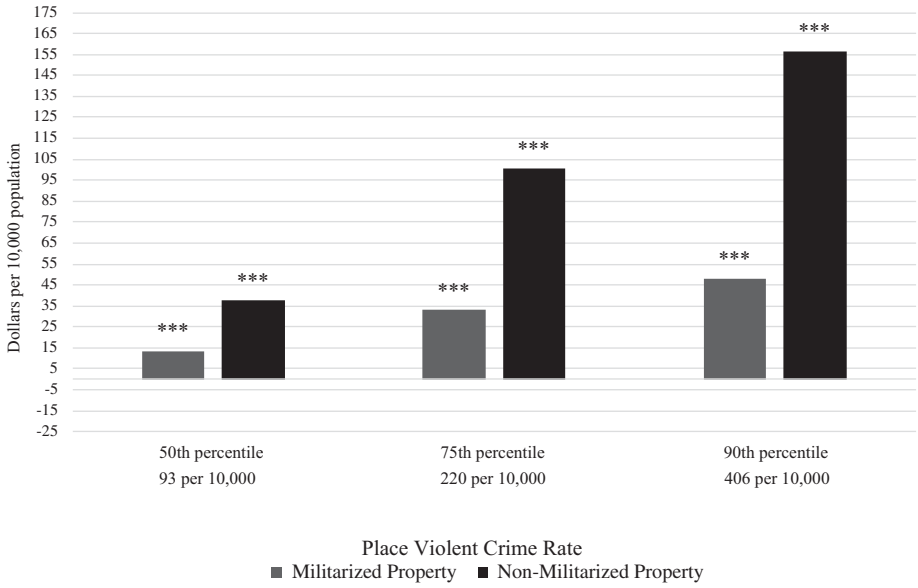
choice arguments for 1033 Program participation, we see a significant and positive relationship between the levels of violent crime an agency reports and participation in the 1033 Program. Agencies with higher violent crime rates are more likely to acquire militarized property from the 1033 Program than agencies with lower violent crime rates. For departments that are successful in acquiring property, however, there is no relationship between crime rates and the value of the property received. Although rates of violent crime are positively associated with 1033 Program participation, county-level drug arrest rates are significantly and negatively associated with participation in the 1033 Program and the value of militarized materiel received by participating agencies. Even though rates of violent crime and drug arrests are associated with 1033 Program participation and the value property received through the 1033 Program, there is no association between recent officer fatalities or the contemporary threat of terror attacks and the acquisition or value of militarized property.

We turn next to our tests of nonlinearity between minority threat and 1033 Program participation. After controlling for local and regional factors and practical concerns associated with policing and 1033 Program participation, we found a curvilinear relationship between the size of the Black population and successfully acquiring *any property* from the 1033 Program. Specifically, the significant and positive coefficient for the linear term and the significant and negative coefficient for the quadratic term indicate that the true relationship is curvilinear, increasing at a decreasing rate. Specifically, as the Black population in a city or town increases, police departments are more likely to participate in the 1033 Program, but the strength of this relationship decreases as the size of the Black population increases. Although there is a curvilinear relationship between the size of the Black population and 1033 Program participation, the relative size of the Hispanic population has no effect on whether a department successfully acquired 1033 Program property. Notably, when we tested the nonlinear relationship between percent Black or percent Hispanic and 1033 Program participation, we found no evidence that this relationship took an exponential form (see appendices E and G in the online supporting information).

For agencies that successfully acquire militarized property through the 1033 Program, we found a nonlinear relationship between the relative size of Black populations in their jurisdictions and the value of equipment they receive each year. The functional form of this relationship for Black minority threat, however, is exponential and increases at an increasing rate. Moreover, although we did not find a nonlinear relationship for Hispanic populations and the acquisition of materiel through the 1033 Program, the relative size of the Hispanic population is significantly associated with the total property value transferred through the 1033 Program. Similar to the Black population, the functional form of this relationship is exponential and increases at an increasing rate. We tested whether the nonlinear relationship between percent Black or percent Hispanic and the value of militarized property received took a quadratic form, but we found no evidence of quadratic functions of minority threat for either Black or Hispanic populations (see appendices F and G in the online supporting information).

In our models for nonmilitarized property in table 3, evidence exists that departments in places with higher local violent crime rates are more likely to acquire nonmilitarized property from the 1033 Program than agencies with lower crime rates, but that crime has no effect on the value of the property received by participating agencies. We find no evidence that county drug arrest rates, police fatalities, or recent terror attacks influence the likelihood of acquiring nonmilitarized equipment via the 1033 Program. Importantly, we

Figure 1. Expected Differences in Property Value (per 10,000 Population) for a 1 Percent Difference in the Violent Crime Rate for Agencies in Places With Violent Crime Rates at the 50th, 75th, and 90th Percentiles



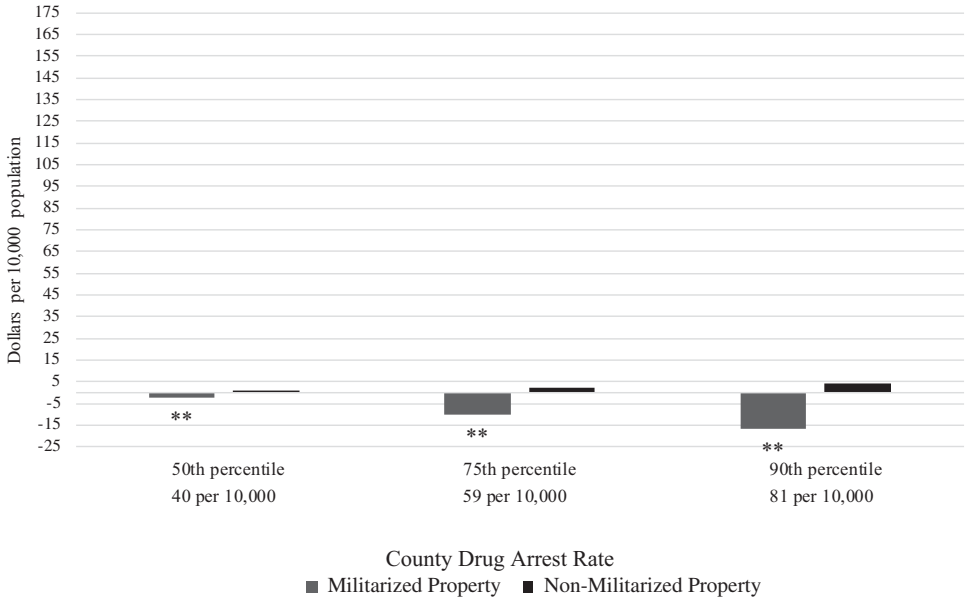
find no evidence that the relative size of minority populations influences the acquisition of nonmilitarized property from the 1033 Program.

Several control variables are associated with receiving both militarized and nonmilitarized property through the 1033 Program. Police departments in places with larger populations are more likely to acquire both militarized and nonmilitarized property successfully through the 1033 Program, but they will receive less in per-capita property if they are successful than agencies in places with smaller populations. Agencies with more officers per capita are more likely to receive militarized and nonmilitarized materiel—and the property they do receive is worth more—than agencies with small police forces. Notably, places with higher unemployment rates are more likely to receive militarized and nonmilitarized materiel, but the relationship between unemployment and militarized property value is not statistically significant.

To facilitate a discussion of the findings, we use the margins command in Stata 15.0 to produce expected differences in the value of acquired militarized and nonmilitarized property (in U.S. dollars) for differences in violent crime rates, drug arrest rates, and the relative sizes of the Black and Hispanic populations, respectively, conditional on the values of each variable and holding all other variables at their means. Because the distribution of crime or arrest rates and our demographic variables are heavily skewed, we present results at their medians and 75th and 90th percentiles to ensure sizable variation across plausible expected values of each variable.

Figure 1 displays the predicted differences in militarized and nonmilitarized property value (per 10,000 population) for a 1 percent difference in the violent crime rate for

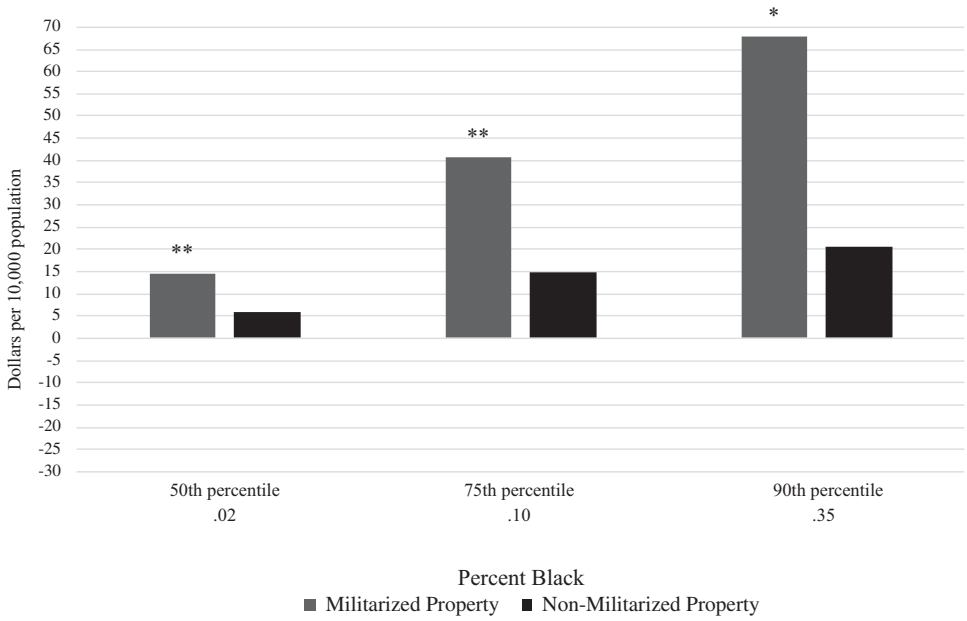
Figure 2. Expected Differences in Property Value (per 10,000 Population) for a 1 Percent Difference in the County Drug Arrest Rate for Agencies in Counties With Drug Arrest Rates at the 50th, 75th, and 90th Percentiles



departments in places with violent crime rates at the national median (93 per 10,000 population), and at the 75th (220 per 10,000 population) and 90th percentiles (406 per 10,000 population). As figure 1 reveals, the relationship between violent crime and the value of militarized and nonmilitarized property received via 1033 Program participation is positive and increases at an increasing rate. For agencies in places with median violent crime rates, a 1 percent difference in the violent crime rate is associated with successfully acquiring about $13 in militarized property and $38 in nonmilitarized property per 10,000 persons. In places with higher violent crime rates, similar differences in violent crime are associated with larger differences in the expected value of 1033 Program property. For example, in places with violent crime rates at the 75th percentile, a 1 percent difference in the violent crime rate is associated with adding approximately $33 in militarized property and $101 in nonmilitarized property per 10,000 persons. In places with extremely high crime rates (top 10th percentile), a 1 percent difference in the crime rate is associated with a difference of about $48 in militarized property per 10,000 persons and $157 in nonmilitarized property from the 1033 Program.

Figure 2 displays the expected differences in militarized and nonmilitarized property value (40 per 10,000 population) for a 1 percent difference in the county drug arrest rates for departments in counties with drug arrest rates at the national median (3 per 10,000 population), and at the 75th (59 per 10,000 population) and 90th percentiles (81 per 10,000 population). As figure 2 reveals, the relationship between county drug arrests and the value of militarized property is negative and decreases at an increasing rate, whereas

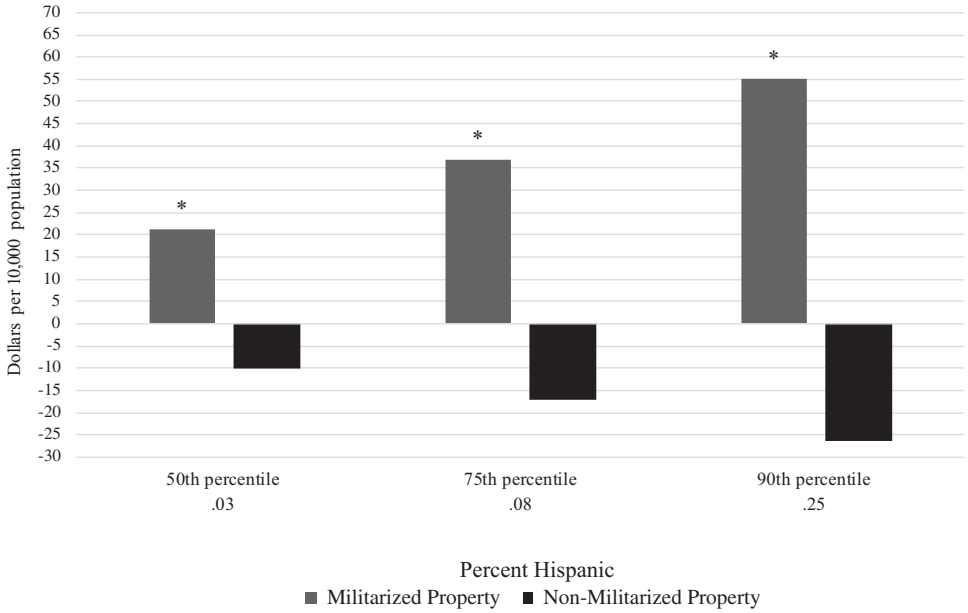
Figure 3. Expected Differences in Property Value (per 10,000 Population) for a 1 Percent Difference in the Black Population for Agencies in Places With Black Population at the 50th, 75th, and 90th Percentiles



the relationship between county drug arrests and the value of nonmilitarized property is positive and increases at an increasing rate. Although statistically significant, differences in county drug arrest rates are associated with small differences in per-capita militarized and nonmilitarized property received. The relationship between county drug arrest rates and the value of property received via 1033 Program participation is small. For example, departments in counties with highest drug arrest rates (e.g., 90th percentile), a 1 percent difference in the county drug arrest rate is associated with $16 less in militarized property per 10,000 persons.

Figure 3 displays the expected differences in militarized and nonmilitarized property value (per 10,000 population) for a 1 percent difference in percent Black for departments in places with Black populations the national median (2 percent), and at the 75th (10 percent) and 90th percentiles (35 percent). As figure 3 reveals, the positive relationship between the relative size of the Black population and the value of militarized and nonmilitarized property received via 1033 Program participation increases at an increasing rate. This relationship, however, is only significant for militarized materiel. For example, for agencies in places with small Black populations (~2 percent), a 1 percent difference in the relative size of the Black population is associated with a difference of about $14 in militarized property per 10,000 persons but only $7 in nonmilitarized property per 10,000 persons. Turning to agencies in places with average Black populations (~10 percent), a 1 percent difference in the Black population is associated

Figure 4. Expected Differences in Property Value (per 10,000 Population) for a 1 Percent Difference in the Hispanic Population for Agencies in Places With Hispanic Population at the 50th, 75th, and 90th Percentiles



with a difference of about $37 in militarized property and $15 in nonmilitarized property per 10,000 persons. In places where Black populations fall in the top 10th percentile (~35 percent), a 1 percent difference in the size of the Black population is associated with a difference of $68 per 10,000 in successfully acquired militarized property but just $21 per 10,000 population worth of nonmilitarized property.

Figure 4 displays the expected differences in militarized and nonmilitarized property value (per 10,000 population) for a 1 percent difference in percent Hispanic for agencies in places with Hispanic populations at the national median (3 percent), at the 75th percentile (8 percent), and at the 90th percentiles (25 percent). Figure 4 reveals that the relationship between the relative size of the Hispanic population and the expected value of militarized property successfully acquired through 1033 Program participation is positive, significant, and increases at an increasing rate. The relationship between the relative size of the Hispanic population and the expected value of nonmilitarized property, however, is negative and insignificant. For example, in places with small Hispanic populations (~3 percent), a 1 percent difference in the relative size of the Hispanic population is associated with a difference of about $21 more per 10,000 persons in militarized property but about $10 less per 10,000 persons in nonmilitarized property. For agencies in places with average Hispanic populations (~8 percent), a similar 1 percent difference in the Hispanic population is associated with successfully acquiring $37 more in militarized property per 10,000 persons. In places with large Hispanic populations (~25 percent), a 1 percent difference in the Hispanic population is associated with a

difference of $55 per 10,000 in successfully acquired militarized property from the 1033 Program.

To summarize, rational choice perspectives of public safety explain 1033 Program participation for the purposes of acquiring both militarized and nonmilitarized property, but they cannot explain differences in the value of total 1033 Program property transferred. Specifically, departments who reported higher violent crime rates or lower drug arrest rates are more likely to acquire militarized and nonmilitarized property from the 1033 Program. Importantly, after considering rational choice explanations, and controlling for several important factors, minority threat variables have a significant and pronounced influence on 1033 Program participation. Moreover, the functional form of minority threat is different in probit models predicting the successful acquisition of property through the 1033 Program and OLS models predicting the value of property transferred to agencies acquiring materiel. Departments in places with larger Black populations are more likely to acquire militarized property through the 1033 Program, but this relationship is curvilinear, increasing at a decreasing rate. For those departments that successfully acquire militarized property, however, there is a nonlinear, exponential relationship between the respective Black and Hispanic populations and the value of property transferred through the 1033 Program that increases at an increasing rate. Finally, the relationship between minority threat variables and 1033 Program participation is significant for militarized materiel but not for nonmilitarized property.

DISCUSSION AND CONCLUSION

In this study, we use a large and unique data set on the value of military equipment obtained by police departments through the Department of Defense's 1033 Program between 1996 and 2016, and we hurdle models to examine the determinants of materiel acquisition by police in the United States. Our results provide support for conventional rational choice arguments claiming that participation in the 1033 Program is a response to higher violent crime rates. Consistent support exists, however, for minority threat arguments that 1033 Program participation is associated with the presence of Black and Hispanic residents in law enforcement jurisdictions for both odds of involvement and the value of materiel acquired. The form of this minority threat varies across the different steps on 1033 Program participation in ways that have significant implications for the policing of minority populations in the United States, yet are consistent with the findings of previous work on criminal justice resource allocation and minority threat (Blalock, 1967; Jackson, 1989; Jackson and Carroll, 1981; Kent and Jacobs, 2005; Stults and Baumer, 2007; Stucky, 2005; Wang and Mears, 2010).

Regarding rational public choice explanations of LEA participation in the 1033 Program, departments with higher violent crime rates were more likely to participate in the Program and received more valuable equipment, both military and nonmilitary, when they did participate. These findings reflect arguments made by proponents of militarization and heads of departments participating in the 1033 Program (Bove and Gavrilova, 2017; Cowper, 2000; Harris et al., 2017). Departments facing high demands to curb violent crime are likely to respond with any technological or procedural changes they believe will help them (Campbell and Campbell, 2009). Interestingly, departments in counties with higher drug arrest rates were less likely to participate in the 1033 Program and received

less valuable materiel when doing so. This may be a result of drug interdiction efforts in counties that produce lower levels of drug arrests seeking out equipment through the 1033 Program to enhance their ability to make arrests, whereas departments in counties with higher arrest rates already have established drug law enforcement capacities that do not perceive a need to seek military equipment (Harris et al., 2017).

As for our other research questions, we find that the relative size of minority populations is consistently and nonlinearly associated with 1033 Program participation. Specifically, the presence of minority populations is associated only with LEA acquisitions of militarized equipment (e.g., weapons, vehicles, and fatigues) from the 1033 Program but *not* of nonmilitarized equipment (e.g., tools and medical supplies). This finding, specific to combat equipment, provides support for the argument that police militarization is, in part, a product of racial threat mechanisms. Specifically, as the relative size of the Black population in places increases, police departments are more likely to acquire surplus military equipment, and as both Black and Hispanic populations increase, the supplies LEAs successfully acquire are worth more money. These findings supplement past studies in which minority populations and formal social control efforts were linked (Jackson, 1989; Kane, 2003; Kent and Jacobs, 2005; Stucky, 2005).

Moreover, our findings add to this literature because we looked for different nonlinear forms of minority threat (Blalock, 1967; Stults and Baumer, 2007; Wang and Mears, 2010).[14] By testing for the effect of quadratic forms of minority presence, we found that LEA selection into 1033 Program *participation* is consistent with a curvilinear minority threat curve, increasing at a decreasing rate, and is only significant for Black populations. This finding may indicate that, controlling for rational public choice demands, LEAs seek surplus military equipment as a response to threats from exclusively Black communities until Blacks make up approximately 27.5 percent of a community. As such, 1033 Program participation seems to be less likely when minority threats are low from Black populations, or once Blacks make up a sizable component of the community. No such finding was found for Hispanic populations, which may indicate that Black populations present greater perceived threats that motivate LEAs to seek materiel.

We similarly found nonlinear effects for minority group presence in the amount of materiel that LEAs successfully acquire. For both Black and Hispanic populations, the amount of 1033 Program combat materiel increased exponentially as anticipated with Blalock's (1967) expectations of a minority power threat curve. Such a finding indicates that although the decision of LEAs to participate in the 1033 Program increases at a decreasing rate with Black populations only, the amount of materiel received increases at an increasing rate for both Black and Hispanic populations. We recognize that the effects sizes displayed in figures 3 and 4 suggest that differences in minority composition do not account for large differences in the expected per-capita value of militarized property across LEAs (e.g., the expected difference in property value for a 1 percent difference is between $14 and $68 in materiel per 10,000 persons). The values displayed in our figures reflect expected differences for all LEAs, however, and not just those participating in the 1033 Program. As such, small differences in the Black populations (e.g., 2.5 percent) are associated with a 1 standard deviation difference in expected militarized property

14. See appendices E through G in the online supporting information for results of other nonlinear forms of minority threat.

value ($169.86 per 10,000; see table 1). Furthermore, it is important to note, for many small agencies, even small differences in the value of militarized items is likely to have a sizable impact on police inventories and local perceptions of law enforcement (Balko, 2013; Brunswick, 2015; Feldman, 2014; Fowler, 2017; Harris et al., 2017). Overall, our findings indicated that LEAs seem to request 1033 Program resources in a pattern that emulates what scholars often find in work on police expenditures (Jackson, 1989; Jackson and Carroll, 1981; Kent and Jacobs, 2005).

Overall, these findings suggest that minority group threat can, in part, account for police acquisition of military equipment. Assuming that 1033 Program participation constitutes an effort of police militarization, this study complements extant work on racial discrimination and the criminal justice system. Racial disparities in policing have roots in systems of racial social control, including slavery and Jim Crow (Alexander, 2012). Additionally, the fact that Black populations, but not Hispanic populations, seem to matter when it comes to initially acquiring militarized property also reflects the fact that White majorities have long viewed large Black populations as a significant economic threat because overt historical institutions of slavery and legal discrimination have been subverted (Alexander, 2012; Du Bois, 1994 [1903]; Hinton, 2016). Once departments successfully acquire property, however, the value of militarized property received is consistent with social control responses in which overwhelming force is used toward both Black and Hispanic minority populations, reflecting a threat-oriented approach rooted in symbolic forms of discrimination (Blalock, 1967). This finding is further bolstered by the fact that minority threat explanations only account for military equipment acquisitions and *not* for noncombat gear from the 1033 Program.

The findings from this research can potentially inform policy decisions about acquisition and use of military equipment by local and state law enforcement. We agree with Balko (2013) that police agencies should make information publicly available on not only the 1033 Program equipment they have acquired but also on how often it is used and in what capacity. Making public data to assess the use (or misuse) of military equipment is wise for a few reasons. First, such disclosures allow the public a means to hold the police accountable for their use of military equipment. Additionally, given the fact that we find that minority presence is significantly associated with 1033 Program participation, a policy of disclosure may help address or illuminate racial biases and tensions that may emerge from police militarization (Meeks, 2006). Finally, it is important to determine whether the acquisition of these materials changes the policing practices of a department (whether for good or for ill; Cowper, 2000; Harris et al., 2017; cf. Buerger, 2000; Kraska, 1996). Accurate data on the acquisition and use of military equipment allows for empirical program evaluations that may inform police departments on the actual effectiveness or ineffectiveness of military equipment used in police work. We also think it is important to acknowledge the role that terrorism and risk to police *do not* play in 1033 Program participation. These research findings show that terrorist attacks and officer fatalities cannot account for participation in the 1033 Program. Although police militarization as a response to the "war on terror" is expected from a rational public choice perspective, we find no support for such a trend in the 1033 Program data.

Even though our results provide support for minority threat accounts of 1033 Program participation and police militarization, several limitations of the current study merit further consideration. First, although we argue that 1033 Program participation is part of the

broader phenomenon of police militarization, we assess only one dimension of that phenomenon. Police militarization goes well beyond participation in the 1033 Program and the acquisition of materiel; it permeates the organizational and cultural environments of police work (Kraska, 2007). Nevertheless, this material dimension of militarization has captured public attention in ways that other aspects of militarization have not. Although the creation of PPUs and their deployments have increased steadily over the past 50 years, recent pictures of police officers riding on MRAPs and other armored vehicles are what make the local and national news outlets (Hutchins, 2014; Welna, 2014). Additionally, the acquisition of military equipment may enable the other dimensions of militarization. For example, surplus vehicles, uniforms, and weapons may foster a combat mentality among officers that encourages overpolicing in a militarized context (Kraska, 1996). The 1033 Program may play a role in this process, but we cannot speak to whether 1033 Program participation affects cultural and organizational dynamics in police work.

Another limitation is that our data cannot speak to LEAs that seek out materiel from private vendors using their own funds or grant monies from federal agencies like the Department of Homeland Security. To illustrate this limitation, consider two hypothetical LEAs that each seek to purchase an armored vehicle. If one agency were to purchase the vehicle from a private vendor rather than using 1033 Program to acquire it, then our data would not capture that transaction. We must be cautious not to equate 1033 Program participation as a uniform indicator of police militarization, as we cannot account for these other private sources of materiel. Similarly, the DLA only includes values for property that has been transferred to law enforcement; thus, our dependent variable only captures successfully acquired property and not all equipment *requested* in a given year.

We maintain, however, that our assessment of police militarization using 1033 Program participation is appropriate for several reasons. The 1033 Program is unique as the value of surplus materiel from the DoD is provided at only shipping cost and with less oversight than comparable approaches that might use grant funds or tax money. We would argue that these incentives for LEAs to use the 1033 Program over private vendors are strong. Furthermore, recent research findings and statements by the DLA indicate that the role of the federal government is limited to the timing of available equipment as a result of year-to-year changes in military engagement globally, geographical location of the storage and the distance of shipping centers to LEAs, agencies' past acquisitions and cumulative value of property accumulated over time, the land area covered by law enforcement, and the ability of LEA administration to complete the application and transfer process successfully (Bruce et al., 2017; Grasso, 2014; Harris et al., 2017). With the exception of the application process itself, we believe that we have controlled for many of these factors in our models.[15] Ultimately, we are interested in examining where materiel is going and whether rational public choice or minority threat accounts of police militarization help to explain the distribution of military equipment. We feel this approach speaks to the

15. Request and cancellation data for the last quarter of 2017 suggest that approximately 50 percent of requests are unsuccessful and that timing and availability of property, agencies' past involvement with the 1033 Program, and LEA mistakes are the most common reasons for rejections. Importantly, if similar equipment is available after the first request is denied, departments are permitted to reapply for any available equipment. Although we are currently in the process of requesting similar data for all years of the program, we believe that an examination of requests and cancellations is beyond the scope of this project and merits important future consideration.

concern of citizens patrolled by police forces using military gear, as well as expands current knowledge about the ecological determinants of police militarization (Ajilore, 2015; Balko, 2013; Bove and Gavrilova, 2017; Bruce et al., 2017; Harris et al., 2017; Insler et al., 2017).

Finally, although traditional measures of racial threat include the relative size of the non-White population, other racial and ethnic dynamics are not considered in this article that likely have important implications for police militarization. For example, Ajilore (2015) found that racial segregation, and not racial composition, was associated with increased odds of acquiring MRAPs. Similarly, race or ethnic-specific measures of economic or political threat may better capture competition for jobs or political power that motivate respective economic or political threat responses. A more comprehensive and causally powerful test of minority threat arguments in police militarization necessitates a theoretical assessment of both racial and ethnic perceptions of threat from the White majority (i.e., is minority threat viewed as political or economic in nature, and does militarization reduce the threat?) and racially specific measures to account for those threats independent of minority presence (e.g., Black and Hispanic employment or voting age population; Shah, 2018; Stults and Baumer, 2007, Wang and Mears, 2010). Our purpose here, however, is to illustrate the importance of race in police militarization arguments and to guide future research in this effort. The presence of significant positive and nonlinear relationships between the Black and Hispanic composition and 1033 Program participation provides early evidence that minority threat and racialized social control may influence police militarization independently of rational public choice accounts.

Future research should be aimed at considering two important questions. First, what are the implications of police militarization in general and participation in the 1033 Program in particular? Alongside the growing attention given to police militarization in the public and the press, there has been increased scrutiny involving police interaction with citizens, particularly young Black males. Whether confrontations and police violence are increasing along with the rise in police militarization remains an unanswered question. Second, although the 1033 Program provides "free" surplus equipment to local law enforcement, the supplies and equipment still come at public cost. In addition to the federally incurred costs of the initial development and purchase of the military equipment, local governments still take on transportation, maintenance, and storage costs, as well additional personnel costs associated with the deployment of this equipment. Scholars should examine how and whether other public services—including welfare programs, infrastructure, and education—are affected by increased social-control spending on military equipment. Furthermore, scholars need to examine whether increases in militarization and declines in other public services may have an interactive effect.

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Appendix A. Federal Supply Category and Class Names for Militarized Materiel Provided by 1033 Program

Federal Supply Category	Federal Supply Class (Examples)
Aircraft/Airframe Structure Compts	Aircraft, Fixed Wing; Aircraft, Rotary Wing
Ammunition And Explosives	Demolition Materials; Land Mines
Clothing/Individual Equipment, Insignia	Armor, Combat Personal; Bag, Weapon Equipment, Carrier; Combat Boots; Specialized Flight Clothing And Accessories
Communication/Detection/Coherent Radiation	Stimulated Coherent Radiation Devices, Components and Accessories; Night Vision and Infrared Goggles; Visible And Invisible Light Communication Equipment
Containers/Packaging/Packing Supplies	Ammunition And Nuclear Ordnance Boxes, Packages And Special Containers
Electrical/Electronic Equipment Components	Antennas, Waveguide, And Related Equipment
Engines And Turbines And Component	Gas Turbines And Jet Engines, Aircraft, Prime Moving; And Components; Gasoline Reciprocating Engines, Aircraft Prime Mover; And Components
Fire Control Equipment	Aircraft Gunnery Fire Control Components; Optical Sighting And Ranging Equipment
Guided Missiles	Guided Missile Handling And Servicing Equipment; Launchers, Guided Missile
Hardware And Abrasives	Hardware, Weapon System
Lighting Fixtures, Lamps	Electric Portable And Hand Lighting Equipment (Crowd Control)
Maintenance/Repair Shop Equipment	Weapons Maintenance And Repair Shop Specialized Equipment; Guided Missile Maintenance, Repair, And Checkout Specialized Equipment; Torpedo Maintenance, Repair, And Checkout Specialized Equipment
Metal Bars, Sheets, Shapes	Armor Plate
Motor Vehicles, Cycles, Trailers	Combat, Assault, And Tactical Vehicles, Tracked; Combat, Assault, And Tactical Vehicles, Wheeled (MRAPs)
Special Industry Machinery	Specialized Ammunition And Ordnance Machinery And Related Equipment
Training Aids And Devices	Armament Training Devices; Kit, Suicide Vest Simulator; Kit, Tactical Vest; Training Aids (Weapons, Counterterrorism)
Vehicular Equipment Components	Adapter Rail, Weapon Mounting; Adapter, Night Vision Viewer; Vehicle Armor Kits; Weapons Systems Specific Vehicular Accessories
Weapons	Assemblies Interchangeable Between Weapons; Camouflage And Deception Equipment; Conversion Kit,5.56 Millimeter Rifle; Guns, Over 30 Mm Up To 75 Mm; Guns, Through 30 Mm; Miscellaneous Weapons; Rifle,5.56 Millimeter
Miscellaneous	Antiterrorism Force Protection Checkpoint Set; Barbed Wire; Battlefield Anti-Intrusion System; Grenade Pouch, Maintenance Of Order And Crowd Control

Appendix B. Federal Supply Category and Class Names for Nonmilitarized Materiel Provided by 1033 Program

Federal Supply Category	Federal Supply Class (Examples)
Adp Equipment/Software/Supplies And Equipment	Central Processing Unit; Components; I/O and Storage Devices; Software; Supplies; System Support Equipment
Agricultural Machinery And Equipment	Gardening Implements And Tool: Harvesting Equipment; Pest, Disease, And Frost Control Equipment; Saddlery, Harness, Whips, And Related Animal Furnishings.
Aircraft Components/Accessories	Aircraft Air Conditioning, Heating, And Pressurizing Equipment; Cargo Tie Down Equipment; Vacuum And De-Icing System Components
Aircraft/Airframe Structure Components	Gliders
Alarm, Signal, Security Detection	Miscellaneous Alarm, Signal, And Security Detection Systems; Shipboard Alarm And Signals Systems; Traffic And Transit Signal Systems
Ammunition And Explosives	Surface Use Explosive Ordnance Disposal Tools And Equipment
Bearings	Mounted and Unmounted
Books, Maps, Other Publications	
Brushes, Paints, Sealers, Adhesives	Paint And Artists' Brushes; Preservative And Sealing Compounds
Chemicals And Chemical Products	Miscellaneous Chemical Specialties; Pest Control Agents And Disinfectants
Cleaning Equipment And Supplies	Brooms, Brushes, Mops, And Sponges; Cleaning And Polishing Compounds And Preparations; Floor Polishers And Vacuum Cleaning Equipment
Clothing/Individual Equipment, Insignia	Badges And Insignia; Backpacks, Noncombat; Hosiery, Handwear, And Clothing Accessories; Luggage; Outerwear; Underwear; Nightwear
Communication/Detection/Coherent Radiation	Accessory Kit, Antenna; Communications Security Equipment And Components; Adaptors; Detector, Explosive Ordnance; Detector, Gas; Intercommunication And Public Address Systems; Headset-Microphone; Miscellaneous Communication Equipment; Radar Equipment; Radio and Television Equipment; Radio Navigation Equipment; Sound Recording And Reproducing Equipment; Video Recording And Reproducing Equipment; Telephone And Telegraph Equipment; Teletype And Facsimile Equipment; X-Ray Equipment
Construction/Mine/Excavate/Highway Equipment	Backhoe, Tractor Mounting; Cranes And Crane-Shovels; Earth Moving And Excavating Equipment; Mining, Rock Drilling, Earth Boring, And Related Equipment; Miscellaneous Construction Equipment; Road Clearing, Cleaning, And Marking Equipment; Truck And Tractor Attachments
Construction And Building Material	Building Components, Prefabricated; Fencing, Fences, Gates And Components; Miscellaneous Construction Materials; Nonwood Construction Lumber And Related Materials; Tile, Brick, And Block; Wallboard, Building Paper, And Thermal Insulation Materials
Containers/Packaging/Packing Supplies	Bags And Sacks; Boxes, Cartons, And Crates; Container, Freight, Thermal; Container, Freight, Utility; Drums And Cans; Packaging And Packing Bulk Materials; Reels And Spools; Specialized Shipping And Storage Containers; Tricon, Freight, General Purpose

Continued

Federal Supply Category	Federal Supply Class (Examples)
Electric Wire, Power Distribution Equipment	Batteries, Nonrechargeable and Rechargeable; Electrical Converters; Electrical Control Equipment; Generators and Generator Sets; Motors, Electrical; Solar Electrical Power Systems; Transformers: Distribution And Power Station; Wire And Cable, Electrical
Electrical/Electronic Equipment Components	Amplifiers; Cable, Cord, And Wire Assemblies: Communication Equipment; Circuit Breakers; Connectors, Electrical; Electrical And Electronic Assemblies; Boards, Cards, And Associated Hardware; Electrical Hardware And Supplies; Electrical Insulators And Insulating Materials; Electronic Modules; Filters And Networks; Fuses, Arresters, Absorbers, And Protectors; Headsets, Handsets, Microphones And Speakers; Lugs, Terminals, And Terminal Strips; Microcircuits, Electronic; Miscellaneous Electrical And Electronic Components; Optoelectronic Devices And Associated Hardware; Relays And Solenoids; Semiconductor Devices And Associated Hardware
Engine Accessories	Engine Air And Oil Filters, Cleaners; Engine Cooling System Components Aircraft Prime Moving; Engine Electrical System Components; Engine Fuel System Components; Miscellaneous Engine Accessories; Turbosuperchargers And Components
Engines And Turbines And Component	Diesel Engines And Components; Gasoline Reciprocating Engines, Except Aircraft; And Components; Miscellaneous Engines And Components
Fiber Optic	Fiber Optic Accessories And Supplies; Fiber Optic Cables; Fiber Optic Kits And Sets
Fire Control Equipment.	Fire Control Computing Sights And Devices; Fire Control Designating And Indicating Equipment; Fire Control Transmitting And Receiving Equipment, Except Airborne; Miscellaneous Fire Control Equipment
Fire/Rescue/Safety; Environmental Protection	Decontaminating And Impregnating Equipment; Fire Fighting Equipment; Hazardous Material Spill Containment And Clean-Up Equipment And Material; Marine Lifesaving And Diving Equipment; Recycling And Reclamation Equipment; Safety And Rescue Equipment
Food Preparation/Serving Equipment	Cutlery And Flatware; Food Cooking, Baking, And Serving Equipment; Kitchen Equipment And Appliances; Kitchen Hand Tools And Utensils; Sets, Kits, Outfits And Modules Food Preparation And Serving; Tableware
Fuels, Lubricants, Oils, Waxes	Oils And Greases: Cutting, Lubricating, And Hydraulic
Furnace/Steam/Drying	Air Purification Equipment; Driers, Dehydrators, And Anhydrators
Furniture	Cabinets, Lockers, Bins, And Shelving; Household Furniture; Miscellaneous Furniture And Fixtures; Office Furniture
Hand Tools	Drill Bits, Counterbores, And Countersinks; Hand And Machine; Hand Tools; Taps, Dies, and Collets; Tool and Hardware Boxes
Hardware And Abrasives	Abrasive Materials; Bolts; Bushings, Rings, Shims And Spacers; Coil, Flat, Leaf, And Wire Springs; Disks And Stones, Abrasive; Fastening Devices; Hardware; Knobs And Pointers; Metal Screening; Nails, Machine Keys, And Pins; Nuts And Washers; Packing And Gasket Materials; Screws; Studs

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Federal Supply Category	Federal Supply Class (Examples)
Household/Commercial Furnishings/Appliances	Floor Coverings; Household And Commercial Utility Containers; Household Furnishings; Ice Making Machine, Cube; Miscellaneous Household And Commercial Furnishings And Appliances
Instruments And Laboratory Equipment	Automatic Pilot Mechanisms And Airborne Gyro Components; Chemical Analysis Instruments; Combination And Miscellaneous Instruments; Drafting, Surveying, And Mapping Instruments; Electrical And Electronic Properties Measuring And Testing Instruments; Engine Instruments; Flight Instruments; Hazard-Detecting Instruments And Apparatus; Laboratory Equipment And Supplies; Liquid And Gas Flow, Liquid Level, And Mechanical Motion Measuring Instruments; Meteorological Instruments And Apparatus; Navigational Instruments; Optical Instruments, Test Equipment, Components And Accessories; Physical Properties Testing And Inspection; Pressure, Temperature, And Humidity Measuring And Controlling Instruments; Scales And Balances; Time Measuring Instruments
Lighting Fixtures, Lamps	Ballasts, Lampholders, And Starters; Electric Lamps; Electric Vehicular Lights And Fixtures; Indoor And Outdoor Electric Lighting Fixtures; Nonelectrical Lighting Fixtures
Lumber, Millwork, Plywood, Veneer	Lumber And Related Basic Wood Materials
Maintenance/Repair Shop Equipment	Fire Control Maintenance And Repair Shop Specialized Equipment; Lubrication And Fuel Dispensing Equipment; Maintenance And Repair Shop Specialized Equipment; Miscellaneous Tools and Adaptors; Motor Vehicle Maintenance And Repair Shop Specialized Equipment Motor Vehicle Maintenance And Repair Shop Specialized Equipment; Miscellaneous Maintenance And Repair Shop Specialized Equipment
Materials Handling Equipment	Blocks, Tackle, Rigging, And Slings; Freight Elevators; Materials Handling Equipment, Not Self-propelled; Miscellaneous Materials Handling Equipment; Warehouse Truck And Tractors, Self-Propelled; Winches, Hoists, Cranes, And Derricks
Measuring Tools	Inspection Gages And Precision Layout Tools; Measuring Tools
Mechanical Power Transmission Equipment	Belting, Drive Belts, Fan Belts, And Accessories; Gears, Pulleys, Sprockets, And Transmission Chain; Miscellaneous Power Transmission Equipment; Torque Converters And Speed Changers
Medical/Dental/Veterinary Equipment/Supplies	Administration Set, Intravenous Fluids; Bag, Medical Instrument And Supply Set; Dental Instruments, Equipment, And Supplies; Hospital And Surgical Clothing And Related Special Purpose Items; Hospital Furniture, Equipment, Utensils, And Supplies; In Vitro Diagnostic Substances, Reagents, Test Kits And Sets; Medical And Surgical Instruments, Equipment, And Supplies; Medicated Cosmetics And Toiletries; Ophthalmic Instruments, Equipment, And Supplies; Replenishable Field Medical Sets, Kits, And Outfits; Surgical Dressing Materials; X-Ray Equipment And Supplies: Medical, Dental, Veterinary
Metal Bars, Sheets, Shapes	Bars And Rods; Plate, Sheet, Strip, And Foil; Structural Shapes, Nonferrous Base Metal

Continued

Federal Supply Category	Federal Supply Class (Examples)
Metalworking Machinery	Accessories For Secondary Metalworking Machinery; Apron, Blacksmiths; Bending And Forming Machines; Cutting And Forming Tools For Secondary Metalworking Machinery; Cutting Tools For Machine Tools; Drilling And Tapping Machines; Electric Arc Welding Equipment; Electric Resistance Welding Equipment; Gas Welding, Heat Cutting, And Metalizing Equipment; Grinding Machines; Hydraulic And Pneumatic Presses, Power Driven; Lathes; Machine Tool Accessories; Machine Tools, Portable; Machining Centers And Way-Type Machines; Manual Presses; Milling Machines; Miscellaneous Machine Tools; Miscellaneous Welding Equipment; Miscellaneous Welding, Soldering, And Brazing Supplies And Accessories; Punching And Shearing Machines; Saws And Filing Machines
Motor Vehicles, Cycles, Trailers	Automobile, Sedan; Camper Top, Truck; Boat, Landing, Inflatable; Boat, Pontoon; Boat, Sport, Inflatable; Ground Effect Vehicles; Motorcycles, Motor Scooters, And Bicycles; Passenger Motor Vehicles; Trailers; Trucks And Truck Tractors, Wheeled
Musical Instruments/Phonograph/Home Radio	Amplifier, Guitar; Musical Instruments; Musical Instrument Parts And Accessories; Phonographs, Radios, And Television Sets: Home-Type
Nonmetallic Fabricated Materials	Glass Fabricated Materials; Miscellaneous Fabricated Nonmetallic Materials; Paper And Paperboard; Plastics Fabricated Materials; Rubber Fabricated Materials
Office Machines	Accounting And Calculating Machines; Miscellaneous Office Machines
Office Supplies And Devices	Office Devices And Accessories; Office Supplies; Stationery And Record Forms
Ores, Minerals And Primary Products	
Photographic Equipment	Cameras, Motion Picture; Cameras, Still Picture; Photographic Equipment And Accessories; Photographic Projection Equipment; Photographic Sets, Kits, And Outfits
Pipe, Tubing, Hose, And Fittings	Hose And Flexible Tubing; Hose, Pipe, Tube, Lubrication, And Railing Fittings; Pipe, Tube And Rigid Tubing
Plumbing, Heating, Waste Disposal	Plumbing Fixtures And Accessories; Space And Water Heating Equipment; Waste Disposal
Prefab Structures/Scaffolding	Bridges, Fixed And Floating; Collective Modular Support System; Miscellaneous Prefabricated Structures; Prefabricated And Portable Buildings; Prefabricated Tower Structures; Rigid Wall Shelters; Scaffolding Equipment And Concrete Forms; Storage Tanks
Pumps And Compressors	Compressors And Vacuum Pumps; Power And Hand Pumps
Recreational/Athletic Equipment	Athletic And Sporting Equipment; Games, Toys, And Wheeled Goods; Recreational And Gymnastic Equipment
Refrigeration, Air Conditioning/Circulation Equipment	Air Conditioning Equipment; Fans, Air Circulators, And Blower Equipment; Refrigeration And Air Conditioning Components; Refrigeration Equipment
Rope, Cable, Chain, Fittings	Chain And Wire Rope; Fiber Rope, Cordage, And Twine; Fittings For Rope, Cable, And Chain

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Federal Supply Category	Federal Supply Class (Examples)
Service And Trade Equipment	Industrial Sewing Machines And Mobile Textile Repair Shops; Laundry And Dry Cleaning Equipment; Miscellaneous Service And Trade Equipment; Vending And Coin Operated Machines; Wrapping And Packaging Machinery
Ship And Marine Equipment	Deck Machinery; Marine Hardware And Hull Items; Miscellaneous Ship And Marine Equipment; Ship And Boat Propulsion Components
Ships, Small Craft, Pontoon, Docks	Miscellaneous Vessels; Small Craft; Food Products Machinery And Equipment
Special Industry Machinery	Miscellaneous Special Industry Machinery; Printing, Duplicating, And Bookbinding Equipment; Pulp And Paper Industries Machinery
Subsistence	Beverages, Nonalcoholic; Composite Food Packages
Textile/Leather/Fur; Tent; Flag	Flags And Pennants; Leather; Notions And Apparel Findings; Tents And Tarpaulins; Textile Fabrics
Tires And Tubes	Tire Rebuilding And Tire And Tube Repair Materials; Tires And Tubes, Pneumatic, Except Aircraft
Toiletries	Personal Toiletry Articles; Toilet Soap, Shaving Preparations, And Dentifrices; Toiletry Paper Products
Tractors	Tractors, Full Tracked, Low Speed; Tractors, Wheeled
Training Aids And Devices	Operation Training Devices
Valves	Valves, Nonpowered
Vehicular Equipment Components	Miscellaneous Vehicular Components; Vehicular Brake, Steering, Axle, Wheel, And Track Components; Vehicular Furniture And Accessories; Vehicular Power Transmission Components
Water Purification/Sewage Treatment	Sewage Treatment Equipment; Water Purification Equipment
Woodworking Machinery And Equipment	Sawmill And Planing Mill Machinery; Tools And Attachments For Woodworking Machinery; Woodworking Machines
Miscellaneous	Absorbent Material, Oil And Water; Adhesives; Collectors And/Or Historical Items; Memorials; Cemeterial And Mortuary Equipment And Supplies; Signs, Advertising Displays, And Identification Plates; Smokers' Articles And Matches

SUPPORTING INFORMATION

Additional supporting information may be found in the online version of this article at the publisher's website.

Appendix C. Correlations for all variables included in analyses (full sample for Probit model, N = 238,013)
Appendix D. Correlations for all variables included in analyses (Agencies currently acquiring militarized property, N = 7,252)
Appendix E. Results from Double-Hurdle Model Estimating 1033 Participation via Acquiring Property and Value of Property Acquired for Militarized and Nonmilitarized Property, 1995–2016 (estimating exponential functional form for acquiring property and property value)
Appendix F. Results from Double-Hurdle Model Estimating 1033 Participation via Acquiring Property and Value of Property Acquired for Militarized and Nonmilitarized

Property, 1995–2016 (estimating quadratic functional form for acquiring property and property value)

Appendix G. Results from Double-Hurdle Model Estimating 1033 Participation via Acquiring Property and Value of Property Acquired for Militarized and Nonmilitarized Property, 1995–2016 (estimating exponential functional form for acquiring property and quadratic functional form for property value)